

Chapter 4

Linear Theory and Stability of Equilibria

In Chapter 2 we discuss how to compute equilibrium solutions to the differential equation $\dot{x} = f(x)$. In this chapter we turn our attention to the behavior of solutions near an equilibrium. This leads to the fundamental notion of stability and the more refined concepts of stable and unstable manifolds, which in turn are most easily identified via the study of linear systems of differential equations. For this reason we start in Section ?? with the theoretical study of homogeneous linear systems

$$\dot{x} = Ax \tag{4.1}$$

where $A \in M_n(\mathbb{R})$. In Section 4.1 we introduce the matrix exponential and prove that it provides an explicit solution unique to (4.1). We use this as an opportunity in Section 4.2 to discuss the concept of a flow and introduce the notion of topological conjugacy. The latter allows us to claim that the behavior of a linear system is determined by the associated eigenvalues and eigenvectors of the vector field.

As suggested above we will make use of the linear theory to gain information about the dynamics of nonlinear dynamics. As is discussed in Chapter 9 the dynamics a nonlinear system in a neighborhood of a hyperbolic equilibrium (the matrix for the linearized system no imaginary eigenvalues) is determined by the dynamics of the linearized system. Therefore, we study the dynamics of hyperbolic linear systems in some detail in Section 4.3 and in Section 4.4 we demonstrate that asymptotic stability for the nonlinear system can be determined from the linear system.

As a consequence it is important to be able to rigorously determine eigenvalues and eigenvectors of matrices, a topic that is the focus of Section 4.5. We end the chapter in Section 4.6, where we introduce the notion of saddle-node bifurcations.

4.1 Existence and Uniqueness of Solutions to Homogeneous Linear Systems

Recall that the solution to the scalar homogeneous linear differential equation $\dot{x} = ax$, $x(0) = x_0 \in \mathbb{R}$ is $x(t) = e^{at}x_0$. We prove an analagous result for (4.1).

Definition 4.1.1. Given $A \in M_n(\mathbb{R})$ define

$$e^A \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

where $A^0 \stackrel{\text{def}}{=} I$ and $0! \stackrel{\text{def}}{=} 1$.

Theorem 4.1.2. For any $A \in M_n(\mathbb{R})$,

$$\sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

is absolutely convergent.

Proof. Let $p = \|A\|$. By Proposition 2.2.2(ii), $\|A^k\| \leq \|A\|^k = p^k$ and therefore

$$\left\| \sum_{k=0}^{\infty} \frac{1}{k!} A^k \right\| \leq \sum_{k=0}^{\infty} \frac{1}{k!} \|A^k\| \leq \sum_{k=0}^{\infty} \frac{1}{k!} p^k = e^p.$$

The same inequalities show that the partial sums form a Cauchy sequence in $M_n(\mathbb{R})$. Since $M_n(\mathbb{R})$ is a Banach space (Proposition 2.2.3) the sequence converges. $\square$

Observe that if $A \in M_n(\mathbb{R})$, then $g(t) \stackrel{\text{def}}{=} e^{tA}$ defines a function $g: \mathbb{R} \rightarrow M_n(\mathbb{R})$.

Proposition 4.1.3.

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A$$

Proof. Using the definition of derivative

$$\begin{aligned} \frac{d}{dt} e^{tA} &= \lim_{h \rightarrow 0} \frac{e^{(t+h)A} - e^{tA}}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{tA} e^{hA} - e^{tA}}{h} \\ &= e^{tA} \lim_{h \rightarrow 0} \frac{e^{hA} - I}{h} \\ &= e^{tA} A \lim_{h \rightarrow 0} \sum_{k=2}^{\infty} \frac{1}{k!} h^{k-1} A^k \\ &= e^{tA} A. \end{aligned}$$

Observe that at the third equality we could have factored as $\lim_{h \rightarrow 0} \left(\frac{e^{hA} - I}{h} \right) e^{tA}$ resulting in $\frac{d}{dt} e^{tA} = A e^{tA}$. $\square$

The following theorem proves the existence and uniqueness of solutions to homogeneous linear systems.

Theorem 4.1.4. *For $A \in M_n(\mathbb{R})$, the unique solution to*

$$\dot{x} = Ax, \quad x(0) = v$$

is

$$x(t) = e^{tA}v.$$

Proof. Let $x(t) = e^{tA}v$. By Proposition 4.1.3 $\dot{x}(t) = A e^{tA}v = Ax$ and $x(0) = v$.

To see that this is the unique solution, let $y(t)$ be any solution. Define $z(t) = e^{-tA}y(t)$. Observe that $z(0) = y(0) = v$ and

$$\frac{dz}{dt} = -A e^{-tA}y(t) + e^{-tA}\dot{y}(t) = -A e^{-tA}y(t) + e^{-tA}Ay(t) = (-A e^{-tA} + e^{-tA}A)y(t) = 0.$$

Therefore,

$$z(t) = e^{-tA}y(t) = v,$$

which implies that $z(t) = e^{tA}v = x(t)$. $\square$

As a consequence of Theorem 4.1.4 we want to be able to compute e^{tA} . For this the following proposition, which is left as an exercise (see Exercise 4.7.1), is useful.

Proposition 4.1.5. *Let $A, B, C \in M_n(\mathbb{R})$. Then:*

(i) *if $A = BCB^{-1}$, then $e^A = Be^CB^{-1}$;*

(ii) *if $AB = BA$, then $e^{A+B} = e^A e^B$;*

(iii) $e^{-A} = (e^A)^{-1}$;

(iv) *if*

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \quad \text{then} \quad e^A = e^a \begin{bmatrix} \cos b & -\sin b \\ \sin b & \cos b \end{bmatrix}.$$

The following lemma is useful for the proof of Proposition 4.1.5(ii).

Lemma 4.1.6. *Let*

$$A = \sum_{k=0}^{\infty} A_k \quad \text{and} \quad B = \sum_{k=0}^{\infty} B_k$$

be absolutely convergent series in $M_n(\mathbb{R})$. Define

$$C = \sum_{k=0}^{\infty} C_k \quad \text{where} \quad C_k \stackrel{\text{def}}{=} \sum_{i+j=k} A_i B_j.$$

Then,

$$C = AB.$$

Proof. A proof by induction shows that

$$\sum_{k=0}^{2m} C_k - \sum_{k=0}^m A_k \sum_{k=0}^m B_k = \sum_{\substack{i+j \leq 2m \\ 0 \leq i \leq m \\ m+1 \leq j \leq 2m}} A_i B_j + \sum_{\substack{i+j \leq 2m \\ m+1 \leq i \leq 2m \\ 0 \leq j \leq m}} A_i B_j$$

Observe that we obtain the proof if we show that this expression converges to 0 as m tends to ∞ .

Using the operator norm

$$\left\| \sum_{k=0}^{2m} C_k - \sum_{k=0}^m A_k \sum_{k=0}^m B_k \right\| \leq \sum_{\substack{i+j \leq 2m \\ 0 \leq i \leq m \\ m+1 \leq j \leq 2m}} \|A_i\| \|B_j\| + \sum_{\substack{i+j \leq 2m \\ m+1 \leq i \leq 2m \\ 0 \leq j \leq m}} \|A_i\| \|B_j\| \quad (4.2)$$

Observe that for the first term

$$\sum_{\substack{i+j \leq 2m \\ 0 \leq i \leq m \\ m+1 \leq j \leq 2m}} \|A_i\| \|B_j\| \leq \left(\sum_{i=0}^{\infty} \|A_i\| \right) \left(\sum_{j=m+1}^{2m} \|B_j\| \right).$$

By assumption

$$\lim_{m \rightarrow \infty} \sum_{i=0}^{\infty} \|A_i\|$$

is bounded and

$$\lim_{m \rightarrow \infty} \sum_{j=m+1}^{2m} \|B_j\| = 0.$$

A similar observation applies to the second term. Therefore,

$$\lim_{m \rightarrow \infty} \left\| \sum_{k=0}^{2m} C_k - \sum_{k=0}^m A_k \sum_{k=0}^m B_k \right\| = 0.$$

□

4.2 Flows and Topological Conjugacy

Given $A \in M_n(\mathbb{R})$ define $\varphi = \varphi_A: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$\varphi(t, x) = e^{tA}x$$

and observe that φ provides the solutions to all initial value problems for $\dot{x} = Ax$. We abstract the essential properties of φ to obtain the following definition.

Definition 4.2.1. Let X be a topological space. A continuous map $\varphi: \mathbb{R} \times X \rightarrow X$ is a *flow* if

- (i) $\varphi(0, x) = x$
- (ii) $\varphi(t, \varphi(s, x)) = \varphi(t + s, x)$

for all $x \in X$ and $t, s \in \mathbb{R}$. The space X is called the *phase space* for the flow. An *orbit through* $x \in X$ is defined by the restriction $\varphi(\cdot, x): \mathbb{R} \rightarrow X$.

Observe that given a linear differential equation $\dot{x} = Ax$ defined on $\mathbb{R}^n$, the function $\varphi: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $\varphi(t, x) := e^{At}x$ is a flow. There are at least two advantages of discussing solutions to differential equations in terms of flows. First, for the typical nonlinear system there is no simple formula for solutions, but nevertheless via the existence theorem (see Theorem 6.1.5) one can derive the existence of a flow (see Theorem 6.2.3). Second, a flow describes all solutions to a differential equation, and hence, provides a language through which the dynamics of different differential equations can be compared.

Definition 4.2.2. Two flows $\varphi: \mathbb{R} \times X \rightarrow X$ and $\psi: \mathbb{R} \times Y \rightarrow Y$ are *topologically equivalent* if there exists a homeomorphism $h: X \rightarrow Y$ such that orbits of φ are mapped onto orbits of ψ preserving the direction of time, that is, there exists a continuous and strictly increasing *time-rescaling map* $\tau: \mathbb{R} \times X \rightarrow \mathbb{R}$ such that

$$h(\varphi(t, x)) = \psi(\tau(t, x), h(x)),$$

for all $(t, x) \in \mathbb{R} \times X$.

Definition 4.2.3. Two flows $\varphi: \mathbb{R} \times X \rightarrow X$ and $\psi: \mathbb{R} \times Y \rightarrow Y$ are *topologically conjugate* if there exists a homeomorphism $h: X \rightarrow Y$ such that

$$h \circ \varphi(t, x) = \psi(t, h(x))$$

for all $t \in \mathbb{R}$ and $x \in X$.

Observe that if two flows are topologically conjugate, then they are topologically equivalent, but the converse need not be true.

Example 4.2.4. Let $\varphi(t, x) = e^{-t}x$ and $\psi(t, x) = e^{-2t}y$ the two flows associated to the problems $\dot{x} = -x$ and $\dot{y} = -2y$, respectively. Define $h : \mathbb{R} \rightarrow \mathbb{R}$ the homeomorphism by

$$h(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

Then, for all $x \in \mathbb{R}$ and $t \in \mathbb{R}$, $h(\varphi(t, x)) = h(e^{-t}x) = e^{-2t}h(x) = \psi(t, h(x))$.

Example 4.2.5. Consider the differential equations

$$\dot{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x \quad \text{and} \quad \dot{x} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} x$$

The resulting flows are topologically equivalent, but not topologically conjugate.

Recall (see e.g. [19, 18]) that the *real Jordan normal form* is a real valued block diagonal matrix

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{bmatrix},$$

where the diagonal blocks J_i , called *Jordan blocks*, take one of the following four forms

$$[\lambda], \quad \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}, \quad \begin{bmatrix} \lambda & 1 & & \\ & \ddots & & \\ & & 1 & \\ & & & \lambda \end{bmatrix}, \quad \begin{bmatrix} D & I & & \\ & \ddots & & \\ & & \ddots & I \\ & & & D \end{bmatrix} \quad (4.3)$$

where I is the 2×2 identity matrix and

$$D = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}.$$

Moreover, any real valued matrix A has a unique, up to the ordering of the diagonal blocks, real Jordan normal form J and there exists an invertible matrix P such that

$$J = P^{-1}AP.$$

Furthermore, $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if J has a Jordan block of the first or third forms in (4.3), and $\alpha \pm i\beta$ are eigenvalues of A if and only if J has a Jordan block of the second or fourth forms in (4.3).

Corollary 4.2.6. *Consider two n -dimensional constant coefficient linear systems*

$$\dot{x} = Ax \quad \text{and} \quad \dot{x} = Bx.$$

If A and B have the same real Jordan normal form then the corresponding flows $\varphi_A(t, x) \stackrel{\text{def}}{=} e^{At}x$ and $\varphi_B(t, x) \stackrel{\text{def}}{=} e^{Bt}x$ are topologically conjugate.

Proof. Since A and B have the same real Jordan normal form (say J), there exist P_1 and P_2 invertible matrices such that

$$P_1^{-1}AP_1 = J = P_2^{-1}BP_2.$$

This implies (using Proposition 4.1.5(i)) that

$$P_1^{-1}e^{At}P_1 = e^{P_1^{-1}AP_1 t} = e^{P_2^{-1}BP_2 t} = P_2^{-1}e^{Bt}P_2,$$

and letting $S \stackrel{\text{def}}{=} P_1P_2^{-1}$ which is invertible, we get

$$e^{Bt} = S^{-1}e^{At}S.$$

Observe that if we set $\varphi(t, x) = \varphi_A(t, x) = e^{At}x$ and $\psi(t, x) = \varphi_B(t, x) = e^{Bt}x$, then

$$S(\psi(t, x)) = \varphi(t, S(x))$$

and hence the two flows are topologically conjugate. $\square$

The following propositions allow us to give an explicit formula for solutions to a homogeneous linear differential equation.

Proposition 4.2.7. *Let $A \in M_n(\mathbb{R})$.*

(i) $\lambda \in \mathbb{R}$ is an eigenvalue of A with associated eigenvector v if and only if $t \mapsto e^{\lambda t}v$ is a solution to (4.1).

(ii) If $\lambda = \alpha + i\beta$ is an eigenvalue of A with associated eigenvector $v = u + iw$, then

$$t \mapsto e^{\alpha t}(\cos(\beta t)u - \sin(\beta t)w) \quad \text{and} \quad t \mapsto e^{\alpha t}(\sin(\beta t)u + \cos(\beta t)w)$$

are solutions to (4.1) and if $\beta \neq 0$, then they are linearly independent.

The proof of Proposition 4.2.7 is left as an exercise.

Proposition 4.2.8. *Let $A = \lambda I + N$ where I denotes the identity matrix and N is nilpotent, i.e. $N^k = 0$, for some $k \geq 1$. Then,*

$$e^{At} = e^{\lambda t} \left(\sum_{j=0}^{k-1} \frac{t^j}{j!} N^j \right).$$

Proof. Since $IN = NI$, by Proposition 4.1.5(ii)

$$e^{At} = e^{(\lambda I + N)t} = e^{\lambda It} e^{Nt} = e^{\lambda t} \left(\sum_{j=0}^{k-1} \frac{t^j}{j!} N^j \right). \quad \square$$

The reader is referred to [18, Chapter 6] for a proof of the following result.

Proposition 4.2.9. *Let A be a $2m \times 2m$ matrix of the form*

$$\begin{bmatrix} D & I & & \\ & & \ddots & \\ & & & I \\ & & & & D \end{bmatrix},$$

where I is the 2×2 identity matrix and $D = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}$. Denoting $R = \begin{bmatrix} \cos(\beta t) & -\sin(\beta t) \\ \sin(\beta t) & \cos(\beta t) \end{bmatrix}$, we have that

$$e^{At} = e^{\alpha t} \begin{bmatrix} R & Rt & Rt^2/2! & \cdots & Rt^{m-1}/(m-1)! \\ R & Rt & Rt^2/2! & \cdots & Rt^{m-2}/(m-2)! \\ & & \ddots & \ddots & \\ & & & R & Rt \\ & & & & R \end{bmatrix}.$$

Given the homogeneous linear differential equation (4.1), let $J = P^{-1}AP$ denote a Jordan normal form for A . Then, since

$$e^{Jt} = \begin{bmatrix} e^{J_1 t} & & & \\ & e^{J_2 t} & & \\ & & \ddots & \\ & & & e^{J_k t} \end{bmatrix},$$

Propositions 4.2.8 and 4.2.9 can be used to express e^{Jt} and by Proposition 4.1.5(i) we have that

$$e^{At} = P e^{Jt} P^{-1}. \quad (4.4)$$

4.3 Hyperbolic Linear Systems

Definition 4.3.1. A linear differential equation $\dot{x} = Ax$ is *hyperbolic* if all the eigenvalues of A have nonzero real parts.

Remark 4.3.2. We will often, in a misuse of terminology, refer to the matrix A as being hyperbolic. As long as it is clear that we are working in the context of differential equations, this is not a problem.

The focus of this section is on characterizing the dynamics of constant coefficient hyperbolic linear systems. We begin by studying the case where all eigenvalues have negative real part.

Theorem 4.3.3. *Consider the differential equation $\dot{x} = Ax$ under the assumption that $A \in M_n(\mathbb{R})$. The following statements are equivalent.*

- (i) *There is a norm $\|\cdot\|_a$ on $\mathbb{R}^n$ and a real number $\lambda > 0$ such that for all $x \in \mathbb{R}^n$ and all $t \geq 0$,*

$$\|e^{At}x\|_a \leq e^{-\lambda t}\|x\|_a.$$

- (ii) *If $\|\cdot\|$ is an arbitrary norm on $\mathbb{R}^n$, then there is a constant $C > 0$ and a real number $\lambda > 0$ such that for all $x \in \mathbb{R}^n$ and all $t \geq 0$,*

$$\|e^{At}x\| \leq Ce^{-\lambda t}\|x\|.$$

- (iii) *The real part of every eigenvalue of A is negative.*

Proof. (i) $\Rightarrow$ (ii) Since all norms on $\mathbb{R}^n$ are equivalent (see Proposition 19.2.4), there exist positive constants K_0 and K_1 such that

$$K_0\|x\| \leq \|x\|_a \leq K_1\|x\|.$$

Therefore,

$$\|e^{At}x\| \leq \frac{1}{K_0}\|e^{At}x\|_a \leq \frac{e^{-\lambda t}}{K_0}\|x\|_a \leq \frac{K_1}{K_0}e^{-\lambda t}\|x\|.$$

(ii) $\Rightarrow$ (iii) The proof is by contradiction. Observe that by (ii) $\lim_{t \rightarrow \infty} \varphi(t, x) = \lim_{t \rightarrow \infty} e^{At}x = 0$, for every $x \in \mathbb{R}^n$. Let $\mu = \alpha + i\beta$ be an eigenvalue of A and assume $\alpha \geq 0$. Let $v = u + iw \neq 0$ be a corresponding eigenvector. By Proposition 4.2.7(ii),

$$t \mapsto e^{\alpha t}(\cos(\beta t)u - \sin(\beta t)w)$$

is a solution. But $\varphi(t, u) = e^{\alpha t}(\cos(\beta t)u - \sin(\beta t)w)$ does not tend to 0 as t goes to infinity, a contradiction.

(iii) $\Rightarrow$ (i) Let $\{v^j \mid j = 1, \dots, n\}$ be a set of linearly independent generalized eigenvectors for A . Let $\nu^j: \mathbb{R} \rightarrow \mathbb{R}^n$ denote the solution to $\dot{x} = Ax$ satisfying $\nu^j(0) = v^j$. Given an initial condition $x = \sum_{j=1}^n c_j v^j$,

$$e^{At}x = \sum_{j=1}^n c_j \nu^j(t).$$

Explicitly writing out (4.4) (that is $e^{At} = Pe^{Jt}P^{-1}$) allows us to conclude that the components of e^{At} can be written as a sum of terms of the form $p(t)e^{\alpha t}\sin(\beta t)$ and $p(t)e^{\alpha t}\cos(\beta t)$, where $\alpha, \beta \in \mathbb{R}$ are the real and imaginary parts of an eigenvalue of A and the polynomial $p(t)$ has degree bounded by the geometric dimension of the corresponding eigenspace. Choose $a > 0$ such that $\alpha < -a$ for all eigenvalues $\alpha + i\beta$ of A . Then, for a given norm $\|\cdot\|$ in $\mathbb{R}^n$, there exists a continuous function $\tau(x)$ such that, for each x

$$\|e^{At}x\| \leq e^{-at}\|x\|$$

for all $t \geq \tau(x)$. By compactness $\tau \stackrel{\text{def}}{=} \sup_{\|x\|=1} \tau(x) < \infty$. Thus, for $\|x\| = 1$, if $t > \tau$, then

$$\|e^{At}x\| \leq e^{-at}.$$

Finally, for arbitrary x

$$\|e^{At}x\| = \|x\| \left\| e^{At} \frac{x}{\|x\|} \right\| \leq \|x\| e^{-at} \quad (4.5)$$

for all $t \geq \tau$.

At this point we have proved that given the norm $\|\cdot\|$ the flow eventually produces a contraction. We claim that

$$\|x\|_a \stackrel{\text{def}}{=} \int_0^\tau e^{as} \|e^{As}x\| ds$$

defines a norm that contracts for all positive time. It is left to the reader to check that $\|\cdot\|_a$ is a norm.

Given $t \geq 0$ choose a positive integer N and $0 \leq T < \tau$ such that $t = N\tau + T$. Observe that

$$\begin{aligned} \|e^{At}x\|_a &= \int_0^\tau e^{as} \|e^{As}e^{At}x\| ds \\ &= \int_0^\tau e^{as} \|e^{A(N\tau+T+s)}x\| ds \\ &= \int_0^{\tau-T} e^{as} \|e^{AN\tau}e^{A(T+s)}x\| ds + \int_{\tau-T}^\tau e^{as} \|e^{A(N+1)\tau}e^{A(T-\tau+s)}x\| ds \\ &\leq \int_0^{\tau-T} e^{a(s-N\tau)} \|e^{A(T+s)}x\| ds + \int_{\tau-T}^\tau e^{a(s-(N+1)\tau)} \|e^{A(T-\tau+s)}x\| ds \\ &= \int_T^\tau e^{a(u-T-N\tau)} \|e^{Au}x\| du + \int_0^T e^{a(u+\tau-T-(N+1)\tau)} \|e^{Au}x\| du \\ &= e^{-at} \int_0^\tau e^{au} \|e^{Au}x\| du \\ &= e^{-at} \|x\|_a, \end{aligned}$$

where the inequality is obtained using (4.5) and we use the substitutions $u = T + s$ in the first integral and $u = T - \tau + s$ in the second integral. Thus (i) holds for the norm $\|\cdot\|_a$. $\square$

The fundamental conclusion of Theorem 4.3.3 is that if every eigenvalue of A has negative real part then every solution to $\dot{x} = Ax$ converges to the origin exponentially fast and furthermore there is a uniform lower bound on the exponential rate.

Denote by $\sigma(A) \subset \mathbb{C}$ the set of all eigenvalues of $A \in M_n(\mathbb{R})$. Given $\lambda \in \sigma(A)$, denote its *generalized eigenspace* by

$$V_\lambda = \{v \in \mathbb{C}^n \mid (A - \lambda I)^n v = 0\}.$$

To extend Theorem 4.3.3 and its conclusions to general hyperbolic linear equations it is useful to introduce the following subspaces.

Definition 4.3.4. The *stable*, *unstable*, and *center eigenspaces* of $\dot{x} = Ax$ for $A \in M_n(\mathbb{R})$ are

$$\begin{aligned}\mathbb{E}_A^s &\stackrel{\text{def}}{=} \text{span} \{u, w \mid u + iw \in V_\lambda, \lambda \in \sigma(A) \text{ and } \text{Re}(\lambda) < 0\} \\ \mathbb{E}_A^u &\stackrel{\text{def}}{=} \text{span} \{u, w \mid u + iw \in V_\lambda, \lambda \in \sigma(A) \text{ and } \text{Re}(\lambda) > 0\} \\ \mathbb{E}_A^c &\stackrel{\text{def}}{=} \text{span} \{u, w \mid u + iw \in V_\lambda, \lambda \in \sigma(A) \text{ and } \text{Re}(\lambda) = 0\}\end{aligned}$$

respectively.

By definition if $\dot{x} = Ax$ is hyperbolic, then $\mathbb{E}_A^c = \{0\}$. This implies that

$$\mathbb{R}^n = \mathbb{E}_A^u \oplus \mathbb{E}_A^s,$$

which is called a *hyperbolic splitting*.

Theorem 4.3.5. Consider the differential equation $\dot{x} = Ax$ where $A \in M_n(\mathbb{R})$.

(i) The subspaces $\mathbb{E}_A^u$, $\mathbb{E}_A^s$ and $\mathbb{E}_A^c$ are invariant under the flow $\varphi(t, x) = e^{At}x$, that is, $\varphi(\mathbb{R}, \mathbb{E}_A^u) = \mathbb{E}_A^u$, $\varphi(\mathbb{R}, \mathbb{E}_A^s) = \mathbb{E}_A^s$ and $\varphi(\mathbb{R}, \mathbb{E}_A^c) = \mathbb{E}_A^c$.

(ii) $x \in \mathbb{E}_A^s$ if and only if there exist $a > 0$ and $C \geq 1$ such that

$$\|e^{At}x\| \leq Ce^{-at}\|x\|, \quad \forall t \geq 0.$$

(iii) $x \in \mathbb{E}_A^u$ if and only if there exist $a > 0$ and $C \geq 1$ such that

$$\|e^{At}x\| \leq Ce^{at}\|x\|, \quad \forall t \leq 0.$$

(iv) $x \in \mathbb{E}_A^c$ if and only if for all $a > 0$

$$\lim_{t \rightarrow \pm\infty} e^{-a|t|}\|e^{At}x\| = 0.$$

Proof. (i) We prove the assertion only for $\mathbb{E}_A^u$. The proofs for the other two subspaces are identical. First note that $\varphi(0, \mathbb{E}_A^u) = \mathbb{E}_A^u$, hence it is sufficient to show that $\varphi(t, \mathbb{E}_A^u) \subset \mathbb{E}_A^u$ for all $t \in \mathbb{R}$. Now, given $\lambda \in \sigma(A)$ and $v \in V_\lambda$, we have that $(A - \lambda I)^n Av = A(A - \lambda I)^n v = 0$, and hence $Av \in V_\lambda$. By induction, $A^k v \in V_\lambda$, for all $k \geq 2$. It then follows that

$$e^{At}v = \lim_{k \rightarrow \infty} \left(I + At + \cdots + \frac{A^k t^k}{k!} \right) v \in V_\lambda.$$

From this we can easily conclude that $\varphi(t, \mathbb{E}_A^u) = e^{At}(\mathbb{E}_A^u) \subset \mathbb{E}_A^u$.

(ii) By (i) $\mathbb{E}_A^s$ is invariant, hence defining $\tilde{A} \stackrel{\text{def}}{=} A|_{\mathbb{E}_A^s}$ we have a linear map $\tilde{A}: \mathbb{E}_A^s \rightarrow \mathbb{E}_A^s$. Notice also that the norm $\|\cdot\|$ on $\mathbb{R}^n$ restricted to $\mathbb{E}_A^s$ is a norm on $\mathbb{E}_A^s$. We then have that $\tilde{A}$ satisfies the conditions of Theorem 4.3.3(iii) and hence the result for $\tilde{A}$ on $\mathbb{E}_A^s$ follows from Theorem 4.3.3(ii), that is, $x \in \mathbb{E}_A^s$ if and only if there exist $a > 0$ and $C \geq 1$ such that

$$\|e^{\tilde{A}t}x\| \leq Ce^{-at}\|x\|, \quad \forall t \geq 0.$$

Finally notice that, for $x \in \mathbb{E}_A^s$, we have that

$$e^{\tilde{A}t}x = e^{At}x$$

and it therefore follows that

$$\|e^{At}x\| = \|e^{\tilde{A}t}x\| \leq Ce^{-at}\|x\|, \quad \forall t \geq 0.$$

(iii) This is essentially the same as (ii) after applying the transformation $t \mapsto -t$.

(iv) If $x \in \mathbb{E}_A^c$, then by construction (4.4) one can see that the growth rate in forward or backward time is subexponential.

So assume $x \notin \mathbb{E}_A^c$. Since $\mathbb{R}^n = \mathbb{E}_A^s \oplus \mathbb{E}_A^u \oplus \mathbb{E}_A^c$, we can write $x = x_s + x_u + x_c$ where $x_* \in \mathbb{E}_A^*$, $*$ $\in \{s, u, c\}$ and $x_s \neq 0$ or $x_u \neq 0$. By (i), (ii) and (iii), $\|x_s(t)\|$ and/or $\|x_u(t)\|$ grows at an exponential rate for negative or positive time, respectively. This contradicts $\lim_{t \rightarrow \pm\infty} e^{-a|t|}\|e^{At}x\| = 0$. $\square$

Theorem 4.3.6. *Let $A, B \in M_n(\mathbb{R})$.*

(i) *If $\dot{x} = Ax$ and $\dot{x} = Bx$ are hyperbolic and the number of eigenvalues with negative real parts are the same for A and B , then the flows e^{At} and e^{Bt} are topologically conjugate.*

(ii) *If $\dot{x} = Ax$ is hyperbolic, then there exists $\epsilon = \epsilon(A)$ such that if $C \in \{M \in M_n(\mathbb{R}) : \|A - M\| < \epsilon\}$, then C is hyperbolic and the flows e^{At} and e^{Ct} are topologically conjugate.*

Proof. First, assume that all eigenvalues of A and B have negative real parts. Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be the norms for $\dot{x} = Ax$ and $\dot{y} = By$ guaranteed by Theorem 4.3.3(i), respectively, that is

$$\|e^{At}x\|_a \leq e^{-\lambda_a t}\|x\|_a, \quad \text{and} \quad \|e^{Bt}y\|_b \leq e^{-\lambda_b t}\|y\|_b,$$

for some $\lambda_a, \lambda_b > 0$ and for all $x, y \in \mathbb{R}^n$ and $t \geq 0$.

Let $S_A \stackrel{\text{def}}{=} \{x \in \mathbb{R}^n \mid \|x\|_a = 1\}$ and $S_B \stackrel{\text{def}}{=} \{y \in \mathbb{R}^n \mid \|y\|_b = 1\}$. Define $h_0: S_A \rightarrow S_B$ by

$$h_0(x) = \frac{x}{\|x\|_b}.$$

Because $\|\cdot\|_a$ is strictly decreasing along the orbits $e^{At}x$ we can define $\tau: \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ to be the solution to

$$e^{A\tau(x)}x \in S_A.$$

Note that

$$\begin{cases} \tau(x) > 0, & \text{if } \|x\|_a > 1 \\ \tau(x) = 0, & \text{if } \|x\|_a = 1 \\ \tau(x) < 0, & \text{if } \|x\|_a < 1. \end{cases}$$

Define $h: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$h(x) := \begin{cases} e^{-B\tau(x)}h_0(e^{A\tau(x)}x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

By construction h is continuous except perhaps at 0. To show continuity at this point, consider a sequence $x_k \rightarrow 0$ and observe that this implies that $\tau(x_k) \rightarrow -\infty$. Since by definition $\|h_0(e^{A\tau(x_k)}x_k)\|_b = 1$, $\lim_{k \rightarrow \infty} h(x_k) = 0 = h(0)$. Thus h is continuous.

That h defines a conjugation follows from the following calculations. If $x = 0$, then

$$h(e^{At}0) = 0 = e^{Bt}0 = e^{Bt}h(0).$$

If $x \neq 0$, then

$$\begin{aligned} h(e^{At}x) &= e^{-B\tau(e^{At}x)}h_0(e^{A\tau(e^{At}x)}e^{At}x) \\ &= e^{-B(\tau(x)-t)}h_0(e^{A(\tau(x)-t)}e^{At}x) \\ &= e^{Bt}e^{-B\tau(x)}h_0(e^{A\tau(x)}e^{-At}e^{At}x) \\ &= e^{Bt}e^{-B\tau(x)}h_0(e^{A\tau(x)}x) \\ &= e^{Bt}h(x). \end{aligned}$$

It is left to the reader to prove that h is a homeomorphism (see Exercise 4.7.5).

Now, assume that the number of negative eigenvalues of A and B are the same. Since A and B are hyperbolic we write them as $A = A_s \oplus A_u$ and $B = B_s \oplus B_u$ by restricting them to their stable and unstable eigenspaces $\mathbb{E}^s$ and $\mathbb{E}^u$, respectively. Assume $\dim \mathbb{E}^s = k$. Let $h_s: \mathbb{R}^k \rightarrow \mathbb{R}^k$ and $h_u: \mathbb{R}^{n-k} \rightarrow \mathbb{R}^{n-k}$ be defined as above using A_s and B_s , and A_u and B_u , respectively. The reader can check that $h := h_s \oplus h_u$ is a conjugacy.

(ii) This follows from Exercise 4.7.6 where it is shown that hyperbolic matrices are generic. $\square$

We can impose a topology $M_n(\mathbb{R})$ by identifying it with $\mathbb{R}^{n^2}$. For example, concatenating the n columns of a matrix to produce a vector of length n^2 . Using this identification define the open sets in $M_n(\mathbb{R})$ to be the pre-images of the open sets in $\mathbb{R}^{n^2}$. Since $\mathbb{R}^{n^2}$ is a metric space, $M_n(\mathbb{R})$ is also a metric space. Since the eigenvalues of matrices are continuous with respect to this topology on $M_n(\mathbb{R})$ (see Theorem 19.7.2), we obtain the following theorem.

Theorem 4.3.7. *The set of hyperbolic matrices is an open subset of $M_n(\mathbb{R})$.*

4.4 Stability of equilibria of nonlinear systems

We now turn our attention to the study of the behaviour of solutions starting near an equilibrium, that is, if $\tilde{x}$ is an equilibrium point we want to know the behaviour of the solutions $\varphi(t, x_0)$ for $\|\tilde{x} - x_0\|$ small. For this we now define the notion of stability.

Definition 4.4.1. An equilibrium $\tilde{x}$ of $\dot{x} = f(x)$ is *stable* if for any $\epsilon > 0$, there exists $\delta > 0$ such that if $\|x_0 - \tilde{x}\| < \delta$, then $\|\varphi(t, x_0) - \tilde{x}\| < \epsilon$ for all $t > 0$. An equilibrium that is not stable is called *unstable*.

A stronger notion of stability is the following.

Definition 4.4.2. An equilibrium $\tilde{x}$ of $\dot{x} = f(x)$ is *asymptotically stable* if it is stable and there exists $\rho > 0$ such that if $\|x_0 - \tilde{x}\| < \rho$, then $\lim_{t \rightarrow \infty} \varphi(t, x_0) = \tilde{x}$.

As the following proposition demonstrates, the stability of the steady state 0 of a constant coefficient linear system $\dot{x} = Ax$ may be determined by the eigenvalues of the matrix A .

Proposition 4.4.3. *Let $A \in M_n(\mathbb{R})$. Then,*

- (i) *If $\operatorname{Re}(\lambda) < 0$ for all $\lambda \in \sigma(A)$, then 0 is asymptotically stable;*
- (ii) *If there exists $\lambda_0 \in \sigma(A)$ such that $\operatorname{Re}(\lambda_0) > 0$, then 0 is unstable.*

Proof. (i) Let $\|\cdot\|$ be an arbitrary norm on $\mathbb{R}^n$. Since $\operatorname{Re}(\lambda) < 0$ for all $\lambda \in \sigma(A)$, then by Theorem 4.3.3, there is a constant $C > 0$ and a real number $\lambda > 0$ such that for all $x \in \mathbb{R}^n$ and all $t \geq 0$,

$$\|e^{At}x\| \leq Ce^{-\lambda t}\|x\|.$$

Let $\epsilon > 0$. Setting $\delta = \epsilon/C$, if $\|x_0\| < \delta$, then for all $t > 0$,

$$\|e^{At}x_0\| \leq Ce^{-\lambda t}\|x_0\| < C\|x_0\| < C\delta = \epsilon.$$

This implies that 0 is a stable equilibrium point. Now, for any $\rho > 0$, $\|x_0\| < \rho$ implies that $\|e^{At}x_0\| \leq Ce^{-\lambda t}\|x_0\| \leq (C\rho)e^{-\lambda t}$ and therefore $\lim_{t \rightarrow \infty} e^{At}x_0 = 0$. Hence, 0 is a (globally)

asymptotically stable equilibrium point.

(ii) Let $\lambda_0 = \alpha + i\beta$ be an eigenvalue of A such that $\alpha \geq 0$, and let $v = u + iw \neq 0$ be a corresponding eigenvector. By Proposition 4.2.7(ii),

$$\varphi_\delta(t, u) \stackrel{\text{def}}{=} \delta e^{\alpha t} (\cos(\beta t)u - \sin(\beta t)w)$$

is a solution, for any $\delta > 0$. Let $\epsilon = 1$. Consider any $\delta > 0$, and let $x_0 = \frac{\delta}{2\|u\|}u$ which satisfies $\|x_0\| < \delta$. Because $\alpha > 0$, there exists $t = t(\delta)$ such that $\|\varphi_\delta(t, x_0)\| \geq \epsilon = 1$. This shows that 0 is not stable, and therefore by definition it is unstable. $\square$

We now turn to the nonlinear problem with the goal of proving Theorem 4.4.6 that states if $\tilde{x}$ is an equilibrium to $\dot{x} = f(x)$, and the real parts of all the eigenvalues of $Df(\tilde{x})$ are negative, then $\tilde{x}$ is asymptotically stable. We begin with a useful characterization of solutions to autonomous perturbations of linear systems.

Proposition 4.4.4. *Let $x(t)$ be a solution of*

$$\dot{x} = Ax + g(t), \quad x(t_0) = x_0. \quad (4.6)$$

Then

$$x(t) = e^{At} \left(e^{-At_0} x_0 + \int_{t_0}^t e^{-As} g(s) ds \right). \quad (4.7)$$

Equation (4.7) is called the *variation of constants* formula.

Proof. Set

$$y(t) \stackrel{\text{def}}{=} e^{-At} x(t)$$

and observe that

$$\dot{y} = -Ae^{-At}x + e^{-At}\dot{x} = -Ae^{-At}x + e^{-At}(Ax + g(t)) = e^{-At}g(t). \quad (4.8)$$

The initial condition $x(t_0) = x_0$ to (4.6) is equivalent to an initial condition $y(t_0) = e^{-At_0}x_0$ to (4.8). We can solve (4.8) by direct integration, i.e.

$$y(t) = e^{-At_0}x_0 + \int_{t_0}^t e^{-As}g(s) ds.$$

Rewriting this in the original coordinates results in equation (4.7). $\square$

The following inequality is fundamental as it controls the rate of growth of solutions to differential equations.

Theorem 4.4.5 (Gronwall's Inequality). *Let $\alpha, \beta: (a, b) \rightarrow [0, \infty)$ be continuous non-negative functions. Assume*

$$\alpha(t) \leq C + \left| \int_{t_0}^t \alpha(s)\beta(s) ds \right|, \quad t_0, t \in (a, b)$$

for some constant $C \geq 0$. Then,

$$\alpha(t) \leq C \exp \left(\left| \int_{t_0}^t \beta(s) ds \right| \right).$$

Proof. Assume first that $a < t_0 \leq t < b$. Define

$$G(t) \stackrel{\text{def}}{=} C + \int_{t_0}^t \alpha(s)\beta(s) ds.$$

Then

$$G'(t) = \frac{dG}{dt}(t) = \alpha(t)\beta(t) \leq G(t)\beta(t).$$

Hence

$$\begin{aligned} \exp \left(- \int_{t_0}^t \beta(s) ds \right) (G'(t) - \beta(t)G(t)) &\leq 0 \\ \frac{d}{dt} \left(G(t) \exp \left(- \int_{t_0}^t \beta(s) ds \right) \right) &\leq 0 \\ G(t) \exp \left(- \int_{t_0}^t \beta(s) ds \right) &\leq G(t_0) \\ G(t) &\leq C \exp \left(\int_{t_0}^t \beta(s) ds \right) \\ \alpha(t) &\leq C \exp \left(\int_{t_0}^t \beta(s) ds \right). \end{aligned}$$

Now assume $a < t \leq t_0 < b$. Define

$$G(t) := C + \int_t^{t_0} \alpha(s)\beta(s) ds$$

and repeat the argument. □

Theorem 4.4.6. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function and let $\tilde{x}$ be an equilibrium solution to*

$$\dot{x} = f(x). \tag{4.9}$$

If all the eigenvalues of $Df(\tilde{x})$ have negative real part strictly less than $-a$, for some $a > 0$, then for any norm $\|\cdot\|$ on $\mathbb{R}^n$ there exist a neighborhood U of $\tilde{x}$ and a constant $C \geq 1$ such that for any initial condition $x_0 \in U$ the solution $\varphi(\cdot, x_0): [0, \infty) \rightarrow \mathbb{R}^n$ satisfies

$$\|\varphi(t, x_0) - \tilde{x}\| \leq Ce^{-at}\|x_0 - \tilde{x}\|, \quad \text{for all } t \geq 0.$$

Therefore $\tilde{x}$ is asymptotically stable.

Proof. We are interested in the dynamics in a neighborhood of $\tilde{x}$, so let $y(t) = x(t) - \tilde{x}$. Observe that using a Taylor approximation (4.9) can be written as

$$\dot{y} = f(\tilde{x} + y) = f(\tilde{x}) + Df(\tilde{x})y + g(y) = Df(\tilde{x})y + g(y) \quad (4.10)$$

where $g(0) = 0$ and $Dg(0) = 0$. Let $y(t)$ be the solution to (4.10) with initial condition $y(0) = y_0$ and view $g(y(t))$ as a function of time. Set $A \stackrel{\text{def}}{=} Df(\tilde{x})$. Using the variation of constants formula (4.7)

$$\begin{aligned} y(t) &= e^{At} \left(y_0 + \int_0^t e^{-As} g(y(s)) ds \right) \\ &= e^{At} y_0 + \int_0^t e^{A(t-s)} g(y(s)) ds. \end{aligned}$$

Since the negative real parts of the eigenvalues of A are strictly less than $-a$, there exists $\lambda = a + \delta > 0$ for some $\delta > 0$ and a constant $C \geq 1$ which satisfy Theorem 4.3.3(ii), that is

$$\|e^{At}x\| \leq Ce^{-\lambda t}\|x\|, \quad (4.11)$$

for all $x \in \mathbb{R}^n$ and all $t \geq 0$. Choose $m > 0$ such that $mC < \delta$. Because $g(0) = 0 \in \mathbb{R}^n$ and $Dg(0) = 0 \in M_n(\mathbb{R})$ there exists $\epsilon > 0$ such that

$$\|g(y)\| \leq m\|y\| \quad \text{for all } \|y\| < \epsilon.$$

By (4.11),

$$\|y(t)\| \leq Ce^{-\lambda t}\|y_0\| + \int_0^t Ce^{-\lambda(t-s)}m\|y(s)\| ds$$

as long as $\|y(s)\| < \epsilon$ for $s \in [0, t]$. Thus,

$$e^{\lambda t}\|y(t)\| \leq C\|y_0\| + \int_0^t Ce^{\lambda s}m\|y(s)\| ds.$$

By Gronwall's inequality (with $\alpha(t) = e^{\lambda t}\|y(t)\|$ and $\beta(t) = Cm$),

$$e^{\lambda t}\|y(t)\| \leq C\|y_0\|e^{Cmt},$$

or equivalently

$$\|y(t)\| \leq C\|y_0\|e^{-(\lambda-Cm)t} < C\|y_0\|e^{-at}. \quad (4.12)$$

Observe that if $\|y_0\| \leq \epsilon/C$, then (4.12) guarantees that $\|y(t)\| < \epsilon$ for all $t > 0$. $\square$

As the following theorem indicates, if at an equilibrium solution the linearized system has an eigenvalue with positive real part, then the equilibrium is unstable. For a proof see [18, Chapter 9.2].

Theorem 4.4.7. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function and let $\tilde{x}$ be an equilibrium solution to $\dot{x} = f(x)$. If there exists an eigenvalue of $Df(\tilde{x})$ with positive real part, then $\tilde{x}$ is unstable.*

4.5 Rigorous Computation of Eigenvalues and Eigenvectors

As is now clear, determining the linear stability of an equilibrium solution requires knowledge of the eigenvalues of a matrix, and determining the stable, unstable, and center eigenspaces requires knowledge of the eigenvectors of a matrix.

In practice, for all but the simplest systems, computation of eigenvalues and eigenvectors must be done numerically and there are a variety of techniques for doing so [3, Chapter 9]. With this in mind we present Theorem 4.5.1 which provides for the existences of solutions (λ, v) of the equation

$$Mv = \lambda v, \quad (4.13)$$

for a given matrix $M \in M_n(\mathbb{C})$ under the assumption that an approximate eigenpair $(\bar{\lambda}, \bar{v})$ has been determined numerically. For the sake of simplicity we do not present how to verify generalized eigenvectors.

Given a vector $x = (x_1, \dots, x_n)$ we denote $x_{\hat{k}} \stackrel{\text{def}}{=} (x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)$, and given a matrix $M \in M_n(\mathbb{C})$, denote by $M_{\hat{k}}$, the $n \times (n-1)$ matrix obtained by deleting the k -th column of M .

Theorem 4.5.1. *Let $\bar{v} \in \mathbb{C}^n$, $\bar{\lambda} \in \mathbb{C}$, and $M \in M_n(\mathbb{C})$. Choose k such that $\bar{v}_k \neq 0$. Define $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ by*

$$f(z) = (M - z_1 I) \begin{bmatrix} z_2 \\ \vdots \\ z_k \\ \bar{v}_k \\ z_{k+1} \\ \vdots \\ z_n \end{bmatrix}. \quad (4.14)$$

Choose $A \in M_n(\mathbb{C})$ and positive constants Y_0 , Z_0 and Z_2 such that

$$\|A(M - \bar{\lambda}I)\bar{v}\| \leq Y_0 \quad (4.15)$$

$$\|I - ADf(\bar{\lambda}, \bar{v}_{\hat{k}})\| \leq Z_0 \quad (4.16)$$

$$2\|A\| \leq Z_2. \quad (4.17)$$

Set $p(r) = Z_2 r^2 - (1 - Z_0)r + Y_0$.

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then A is invertible and there exists unique $\tilde{v}$ and $\tilde{\lambda}$ such that $\tilde{v}_k = \bar{v}_k$ and $\|(\tilde{\lambda}, \tilde{v}) - (\bar{\lambda}, \bar{v})\| < r_0$, and

$$M\tilde{v} = \tilde{\lambda}\tilde{v}.$$

Perhaps a few words of motivation are in order before turning to the details of the proof. Theorem 4.5.1 is a special case of Theorem 2.4.2. In particular, we can rewrite (4.13) as

$$g(\lambda, v) = (M - \lambda I)v = 0.$$

This turns the problem of finding an eigenpair into the problem of finding a zero of g which suggests that we consider the fixed point problem

$$T(z) = z - Ag(z).$$

However, we cannot hope to use a radii polynomial to find a fixed point of T . To understand why, observe that given a solution $(\tilde{\lambda}, \tilde{v})$ of (4.13) and any $\theta \in \mathbb{C} \setminus \{0\}$, $(\tilde{\lambda}, \theta\tilde{v})$ is also a solution. This implies that the solution $(\tilde{\lambda}, \tilde{v})$ is not *isolated*, that is there is no neighborhood of $(\tilde{\lambda}, \tilde{v})$ in $\mathbb{C}^{n+1}$ on which $(\tilde{\lambda}, \tilde{v})$ is the unique solution to $g(\tilde{\lambda}, \tilde{v}) = 0$. To address this issue, we introduce the notion of a *phase condition* which will ensure that solutions are isolated. Notice that uniqueness fails along a one dimensional parameter space $\mathbb{C} \setminus \{0\}$, when viewed as a subset of the one dimensional complex vector space $\mathbb{C}$. Thus, one expects that if we can find an appropriated reduction by one of the (complex) dimensions of the space on which g acts, then the reduced function will have a unique fixed point. With this in mind we choose a phase condition that involves fixing one of the components of v to be a given constant. We could also reinterpret $\mathbb{C}$ as a two dimensional real vector space, and see $\mathbb{C} \setminus \{0\}$ as a two dimensional parameter space. In this case one would need two additional (real) constraints. In particular when computing real eigenvalues and eigenvectors we just need to add one additional real constraint, such as $\|v\|^2 = 1$ for example (see Exercise 4.7.8).

To be more precise, suppose that an approximate eigenpair $(\bar{\lambda}, \bar{v})$ of M has been computed, i.e., $M\bar{v} \approx \bar{\lambda}\bar{v}$ (note that this is the first line of the hypothesis of Theorem 4.5.1). Choose k such that $v_k \neq 0$ and define $f: \mathbb{C}^n \rightarrow \mathbb{C}^n$ by (4.14). By definition, a solution z of $f(z) = 0$ corresponds to an eigenpair (λ, v) of M with the eigenvalue $\lambda = z_1$ and the eigenvector $v = (z_2, \dots, z_k, \bar{v}_k, z_{k+1}, \dots, z_n)$. Since the k -th coordinate of v is fixed and there are finitely many eigenvectors up to scalar multiplication, if (λ, v) is an eigenpair, then it is isolated as a point in $\mathbb{C}^{n+1}$.

Proof of Theorem 4.5.1. To proof follows as an application of Theorem 2.4.2 where f is given by (4.14). Observe that $f(\bar{\lambda}, \bar{v}_k) = (M - \bar{\lambda}I)\bar{v}$, hence satisfying (4.15) implies that (2.14) is satisfied. Similarly, (4.16) is a restatement of (2.15). Thus we turn our attention to showing that (4.17) verifies (2.16).

Observe that

$$Df(z) = \left[\begin{array}{c|c} \begin{matrix} -z_2 \\ \vdots \\ -z_k \\ -\bar{v}_k \\ -z_{k+1} \\ \vdots \\ -z_n \end{matrix} & (M - z_1 I)_{\widehat{k}} \end{array} \right].$$

Therefore,

$$Df(c) - Df(\bar{\lambda}, \bar{v}_{\hat{k}}) = \left[\begin{array}{c|c} \begin{matrix} -c_2 + \bar{v}_2 \\ \vdots \\ -c_k + \bar{v}_{k-1} \\ 0 \\ -c_{k+1} + \bar{v}_{k+1} \\ \vdots \\ -c_n + \bar{v}_n \end{matrix} & ((-c_1 + \bar{\lambda})I)_{\widehat{k}} \end{array} \right]$$

We are interested in $c \in \overline{B_r((\bar{\lambda}, \bar{v}_{\hat{k}}))}$ or equivalently $c \in (\bar{\lambda} + z_1, \bar{v}_{\hat{k}} + z_1)$ where $\|z\| \leq r$. Thus

$$\|Df(c) - Df(\bar{\lambda}, \bar{v}_{\hat{k}})\| = \left\| \left[\begin{array}{c|c} \begin{matrix} -z_2 \\ \vdots \\ -z_k \\ 0 \\ -z_{k+1} \\ \vdots \\ -z_n \end{matrix} & (-z_1 I)_{\widehat{k}} \end{array} \right] \right\| \leq 2\|z\| \leq 2r.$$

Therefore,

$$\|A[Df(c) - Df(\bar{\lambda}, \bar{v}_{\hat{k}})]\| \leq 2\|A\|r,$$

and hence (4.17) verifies (2.16). The conclusion now follows from Theorem 2.4.2. $\square$

Remark 4.5.2. In practice, when choosing a k such that $\bar{v}_k \neq 0$ in Theorem 4.5.1, we choose k such that

$$|\bar{v}_k| = \max \{|\bar{v}_j| \mid j = 1, \dots, n\}.$$

The reason being that the neighborhood where we have a unique solution cannot contain the origin, hence $|\bar{v}_k|$ very small forces this neighborhood to be very small.

Example 4.5.3. Recall that the Lorenz system is given by

$$\begin{cases} \dot{x}_1 = \sigma(x_2 - x_1) \\ \dot{x}_2 = \rho x_1 - x_2 - x_1 x_3 \\ \dot{x}_3 = -\beta x_3 + x_1 x_2. \end{cases}$$

For any $\beta > 0$ and $\rho > 1$, the set of equilibria of (2.27) is given by

$$\left\{ (0, 0, 0), \left(\pm\sqrt{\beta(\rho-1)}, \pm\sqrt{\beta(\rho-1)}, \rho-1 \right) \right\}.$$

The stability of each hyperbolic equilibrium solution $x = (x_1, x_2, x_3)$ is determined by the eigenvalues of $Df(x)$. For instance, at the *positive eye* $x = \left(\sqrt{\beta(\rho-1)}, \sqrt{\beta(\rho-1)}, \rho-1 \right)$, the Jacobian matrix is given by

$$Df(x) = \begin{pmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & -\sqrt{\beta(\rho-1)} \\ \sqrt{\beta(\rho-1)} & \sqrt{\beta(\rho-1)} & -\beta \end{pmatrix}.$$

For the classical parameter values $\sigma = 10$, $\beta = \frac{8}{3}$ and $\rho = 28$, numerical calculations lead to the following candidates for eigenvalues:

$$\bar{\lambda}_1 = -13.854577914596042, \quad \bar{\lambda}_{2,3} = 0.093955623964686 \pm 10.194505220927850i$$

and their associated eigenvectors

$$\bar{v}^1 = \begin{pmatrix} 0.855665024602210 \\ -0.329822750612395 \\ -0.398816146677856 \end{pmatrix}, \quad \bar{v}^{2,3} = \begin{pmatrix} -0.266119316830765 \pm 0.295010166256352i \\ 0.032128610535763 \pm 0.569077429163104i \\ -0.719213558699417 \end{pmatrix}.$$

Consider the approximation $(\bar{\lambda}_1, \bar{v}^1)$. The component of $\bar{v}^1$ with largest magnitude is $\bar{v}_1^1$. Defining the radii polynomial as

$$p(r) = Z_2 r^2 - (1 - Z_0)r + Y_0,$$

where Y_0 is defined by (4.15), and Z_0 and Z_2 are given by (4.16) and (4.17) respectively, we obtained an interval of existence $I = [9.3438 \times 10^{-15}, 0.64912]$.

Here it is worth to note that the large radius $r_+ = 0.64912$ above does not exclude the existence of nearby eigenvalues. This radius gives a neighborhood where we have uniqueness of the eigenpair $(\bar{\lambda}, \bar{v})$. We could, however, have a nearby eigenvalue with an eigenvector that is outside of the neighborhood for $\bar{v}$.

For $\lambda_{2,3}$ we obtain the interval of existence $[5.6641 \times 10^{-15}, 0.45322]$. This altogether yields a proof that the true eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $Df(x)$ satisfy

$$\lambda_1 < 0 < \operatorname{Re}(\lambda_2) = \operatorname{Re}(\lambda_3)$$

and therefore that the positive eye $x = \left(\sqrt{\beta(\rho-1)}, \sqrt{\beta(\rho-1)}, \rho-1 \right)$ is an unstable equilibrium solution.

As discussed in Section 4, analyzing the stability of an equilibrium solution $\tilde{x}$ necessitates computing eigenvalues of $Df(\tilde{x})$. In Example 2.4.9 this was done using the exact formula for the equilibrium to compute $Df(\tilde{x})$. However in general we will not know the equilibria exactly, and hence we need to compute the eigenvalues without knowing $Df(\tilde{x})$ exactly. One way to do this is to compute the equilibrium and the eigenpairs at the same time as described next.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and suppose we want to compute an equilibrium solution $\tilde{x}$ and s eigenpairs of $Df(\tilde{x})$. Assume that an approximated equilibrium solution $\bar{x}$ and s approximated eigenpairs $(\bar{\lambda}^i, \bar{v}^i)$, $i = 1, \dots, s$, of $Df(\bar{x})$ have been computed. For each $i = 1, \dots, s$, choose k_i such that $\bar{v}_{k_i}^i \neq 0$ and define the map $g_i : \mathbb{R}^n \times \mathbb{C}^n \rightarrow \mathbb{C}^n$, corresponding to (4.14), given by

$$g_i(x, z) := (Df(x) - z_1 I) \begin{bmatrix} z_2 \\ \vdots \\ z_{k_i} \\ \bar{v}_{k_i}^i \\ z_{k_i+1} \\ \vdots \\ z_n \end{bmatrix},$$

and consider the map $F : \mathbb{R}^n \times \mathbb{C}^n \times \dots \times \mathbb{C}^n \rightarrow \mathbb{R}^n \times \mathbb{C}^n \times \dots \times \mathbb{C}^n$ given by

$$\begin{bmatrix} x \\ z^1 \\ \vdots \\ z^s \end{bmatrix} \mapsto F \begin{bmatrix} x \\ z^1 \\ \vdots \\ z^s \end{bmatrix} = \begin{bmatrix} f(x) \\ g_1(x, z^1) \\ \vdots \\ g_s(x, z^s) \end{bmatrix}. \quad (4.18)$$

Applying the radii polynomial approach to compute a zero of F will provide the equilibrium solution $\tilde{x}$ of $\dot{x} = f(x)$ and the s eigenpairs of $Df(\tilde{x})$ as shown in the next theorem.

Theorem 4.5.4. *Let $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, $\bar{x} \in \mathbb{R}^n$, $\bar{v}^i \in \mathbb{C}^n$, and $\bar{\lambda}^i \in \mathbb{C}$, for $i = 1, \dots, s$. Define $F : \mathbb{R}^n \times \mathbb{C}^n \times \dots \times \mathbb{C}^n \rightarrow \mathbb{R}^n \times \mathbb{C}^n \times \dots \times \mathbb{C}^n$ by (4.18) and set $\bar{z} = (\bar{x}, \bar{\lambda}^1, \bar{v}_{k_1}^1, \dots, \bar{\lambda}^s, \bar{v}_{k_s}^s)$. Choose $A \in M_{(s+1)n}(\mathbb{C})$ and positive constants Y_0 , Z_0 and Z_2 such that*

$$\|AF(\bar{z})\| \leq Y_0 \quad (4.19)$$

$$\|I - ADF(\bar{z})\| \leq Z_0 \quad (4.20)$$

$$\|A[DF(c) - DF(\bar{z})]\| \leq Z_2(r)r, \quad \text{for all } c \in \overline{B_r(\bar{z})} \text{ and all } r > 0. \quad (4.21)$$

Define $p(r) = Z_2(r)r^2 - (1 - Z_0)r + Y_0$.

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then A is invertible and there exists unique $\tilde{x}$, $\tilde{v}^i$, and $\tilde{\lambda}^i$ such that $\tilde{v}_{k_i}^i = \bar{v}_{k_i}^i$ for $i = 1, \dots, s$, and $\|(\tilde{x}, \tilde{\lambda}^1, \tilde{v}^1, \dots, \tilde{\lambda}^s, \tilde{v}^s) -$

$(\bar{x}, \bar{\lambda}^1, \bar{v}^1, \dots, \bar{\lambda}^s, \bar{v}^s) \| < r_0$, and

$$\begin{aligned} f(\tilde{x}) &= 0 \\ Df(\tilde{x})\tilde{v}^i &= \tilde{\lambda}^i \tilde{v}^i, \quad \text{for } i = 1, \dots, s. \end{aligned}$$

Proof. Note that, analogously to Theorem 4.5.1, the proof is a direct application of Theorem 2.4.2 to (4.18). Hence we only indicate how to compute the derivative in (4.21). We have that

$$DF(x, z^1, \dots, z^s) = \begin{bmatrix} D_x f(x) & 0 & 0 & \cdots & 0 \\ D_x g_1(x, z^1) & D_{z^1} g_1(x, z^1) & 0 & \cdots & 0 \\ D_x g_2(x, z^2) & 0 & D_{z^2} g_2(x, z^2) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ D_x g_s(x, z^s) & 0 & 0 & \cdots & D_{z^s} g_s(x, z^s) \end{bmatrix}$$

where

$$D_{z^i} g_i(x, z^i) = \left[\begin{array}{c|c} -z_2^i & \\ \vdots & \\ -z_{k_i}^i & \\ -\bar{v}_{k_i}^i & \\ -z_{k_i+1}^i & \\ \vdots & \\ -z_n^i & \end{array} \right] (D_x f(x) - z_1^i I)_{k_i}^{\wedge}. \quad \square$$

Theorem 4.5.4 provides an alternative to compute the eigenpairs when the equilibrium is unknown by computing the equilibrium and the eigenpairs at the same time. However if one has already computed the equilibria and have error bounds on the computed equilibria and/or estimates on the derivatives of f , one may wish to compute the eigenpairs using this information.

One possibility is to incorporate the error bounds on the Jacobian matrix and define a interval valued matrix containing the Jacobian of interest and compute the eigenpairs of this matrix (see Exercise 4.7.10).

Another alternative, assuming we have rigorously verified the existence of an equilibria $\tilde{x}$ in some ball $\overline{B_{r_*}(\tilde{x})}$, and assuming we have bounds on $D^2 f$ on this ball, is given by the next theorem.

Theorem 4.5.5. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 map, $\tilde{x}, \bar{x} \in \mathbb{R}^n$, and assume that there is a $r_* > 0$ such that $\|\bar{x} - \tilde{x}\| \leq r_*$. Let also $\bar{v} \in \mathbb{C}^n$, and $\bar{\lambda} \in \mathbb{C}$. Choose k such that $\bar{v}_k \neq 0$.*

Define $g: \mathbb{C}^n \rightarrow \mathbb{C}^n$ by

$$g(z) = (Df(\tilde{x}) - z_1 I) \begin{bmatrix} z_2 \\ \vdots \\ z_k \\ \bar{v}_k \\ z_{k+1} \\ \vdots \\ z_n \end{bmatrix}. \quad (4.22)$$

Choose $A \in M_n(\mathbb{C})$ and positive constants $\tilde{Y}_0$, $\tilde{Z}_0$ and Z_2 such that

$$\|A(Df(\bar{x}) - \bar{\lambda}I)\bar{v}\| \leq \tilde{Y}_0 \quad (4.23)$$

$$\|I - A[-\bar{v}](Df(\bar{x}) - \bar{\lambda}I)_{\hat{k}}\| \leq \tilde{Z}_0 \quad (4.24)$$

$$2\|A\| \leq Z_2. \quad (4.25)$$

Suppose there is a constant $\eta > 0$ such that $\|D^2f(c)\| \leq \eta$ for all $c \in \overline{B_{r_*}(\bar{x})}$. Define $Y_0 \stackrel{\text{def}}{=} \tilde{Y}_0 + \eta\|A\|\|\bar{v}\|r_*$, $Z_0 \stackrel{\text{def}}{=} \tilde{Y}_0 + \eta\|A\|r_*$, and set $p(r) = Z_2r^2 - (1 - Z_0)r + Y_0$.

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then A is invertible and there exists unique $\tilde{v}$ and $\tilde{\lambda}$ such that $\|(\tilde{\lambda}, \tilde{v}) - (\bar{\lambda}, \bar{v})\| < r_0$, and

$$Df(\tilde{x})\tilde{v} = \tilde{\lambda}\tilde{v}.$$

Proof. The proof is another application of Theorem 2.4.2, this time to the function g given by (4.22). First we show that satisfying (4.23) implies that (2.14) is satisfied. Setting $\bar{z} \stackrel{\text{def}}{=} (\bar{\lambda}, \bar{v}_{\hat{k}})$, we have

$$\begin{aligned} \|Ag(\bar{z})\| &= \|A(Df(\tilde{x}) - \bar{\lambda}I)\bar{v}\| \leq \|A(Df(\bar{x}) - \bar{\lambda}I)\bar{v}\| + \|A(Df(\tilde{x}) - Df(\bar{x}))\bar{v}\| \\ &\leq \tilde{Y}_0 + \|A\| \left(\sup_{c \in \overline{B_{r_*}(\bar{x})}} \|D^2f(c)\| \right) \|\tilde{x} - \bar{x}\|\|\bar{v}\| \leq \tilde{Y}_0 + \eta\|A\|\|\bar{v}\|r_* = Y_0. \end{aligned}$$

Next if (4.24) is satisfied, recalling that

$$Dg(z) = \left[\begin{array}{c|c} \begin{matrix} -z_2 \\ \vdots \\ -z_k \\ -\bar{v}_k \\ -z_{k+1} \\ \vdots \\ -z_n \end{matrix} & (Df(\tilde{x}) - z_1 I)_{\hat{k}} \end{array} \right],$$

we have

$$\begin{aligned}
\|I - ADg(\bar{z})\| &= \|I - A[-\bar{v}|(Df(\tilde{x}) - \bar{\lambda}I)_{\widehat{k}}]\| \\
&\leq \|I - A[-\bar{v}|(Df(\bar{x}) - \bar{\lambda}I)_{\widehat{k}}]\| + \|A[0|(Df(\tilde{x}) - Df(\bar{x}))_{\widehat{k}}]\| \leq \\
&\leq \|I - A[-\bar{v}|(Df(\bar{x}) - \bar{\lambda}I)_{\widehat{k}}]\| + \|A\| \|Df(\tilde{x}) - Df(\bar{x})\| \leq \\
&\leq \tilde{Z}_0 + \|A\| \left(\sup_{c \in B_{r_*}(\bar{x})} \|D^2f(c)\| \right) \|\tilde{x} - \bar{x}\| \leq \tilde{Y}_0 + \eta \|A\| r_* = Z_0,
\end{aligned}$$

and therefore (2.15) is satisfied.

Finally, as in the proof of Theorem 4.5.1, we have that

$$Dg(c) - Dg(\bar{z}) = \begin{bmatrix} -c_2 + \bar{v}_2 \\ \vdots \\ -c_k + \bar{v}_{k-1} \\ 0 \\ -c_{k+1} + \bar{v}_{k+1} \\ \vdots \\ -c_n + \bar{v}_n \end{bmatrix} ((-c_1 + \bar{\lambda})I)_{\widehat{k}}.$$

It then follows that

$$\|Dg(c) - Dg(\bar{z})\| \leq 2r,$$

and therefore,

$$\|A[Dg(c) - Dg(\bar{z})]\| \leq 2\|A\|r.$$

Hence (4.25) verifies (2.16). The conclusion now follows from Theorem 2.4.2. $\square$

Example 4.5.6. Consider again the Lorenz system

$$\begin{cases} \dot{x}_1 = \sigma(x_2 - x_1) \\ \dot{x}_2 = \rho x_1 - x_2 - x_1 x_3 \\ \dot{x}_3 = -\beta x_3 + x_1 x_2, \end{cases}$$

with the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$. We will consider the same equilibrium solution computed in Example 2.4.9, but this time rather than using the explicit formulas, we will compute the equilibrium and the eigenpairs using Theorem 4.5.4 to illustrate how to proceed on more realistic problems when we do not have explicit formulas for the solutions.

Numerical computations provide the following approximations for the equilibrium

$$\bar{x} = (8.4852813742386, 8.4852813742386, 27.0),$$

for eigenvalues of the associated Jacobian

$$\bar{\lambda}_1 = -13.854577914596042, \quad \bar{\lambda}_{2,3} = 0.093955623964686 \pm 10.194505220927850i,$$

and their associated eigenvectors

$$\bar{v}^1 = \begin{pmatrix} 0.855665024602210 \\ -0.329822750612395 \\ -0.398816146677856 \end{pmatrix}, \quad \bar{v}^{2,3} = \begin{pmatrix} -0.266119316830765 \pm 0.295010166256352i \\ 0.032128610535763 \pm 0.569077429163104i \\ -0.719213558699417 \end{pmatrix}.$$

Using this approximations we define the radii polynomials as

$$p(r) = Z_2 r^2 - (1 - Z_0)r + Y_0,$$

where Y_0 , Z_0 , and Z_2 are given by (4.19), (4.20), and (4.21) respectively, and apply Theorem 4.5.4 to obtain the interval of existence $I = [7.8374 \times 10^{-14}, 0.3872]$.

Again, as in Example 2.4.9, we get that the true eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $Df(\tilde{x})$ satisfy

$$\lambda_1 < 0 < \operatorname{Re}(\lambda_2) = \operatorname{Re}(\lambda_3)$$

and therefore that the equilibrium $\tilde{x}$ is unstable.

4.6 Saddle-Node Bifurcation

The equilibria for the differential equation $\dot{x} = f(x, \lambda)$, $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ are solutions to $f(x, \lambda) = 0$. By the implicit function theorem, if $f(\tilde{x}, \tilde{\lambda}) = 0$ and $D_x f(\tilde{x}, \tilde{\lambda})$ is invertible then there exists a smooth family of equilibria $\tilde{x}(\lambda)$ defined in a neighborhood of $\tilde{\lambda}$. Denote $A_\lambda \stackrel{\text{def}}{=} D_x f(\tilde{x}(\lambda), \lambda)$, and assume it is hyperbolic, that is $\sigma(A_\lambda) \cap i\mathbb{R} = \emptyset$. By continuity of $\tilde{x}(\lambda)$ and $D_x f$, there exists $\delta > 0$ such that $\sigma(A_\lambda) \cap i\mathbb{R} = \emptyset$ for all $\lambda \in B_\delta(\tilde{\lambda}) = (\tilde{\lambda} - \delta, \tilde{\lambda} + \delta)$. In other words, for any $\lambda_0, \lambda_1 \in B_\delta(\tilde{\lambda})$, A_{λ_0} and A_{λ_1} are hyperbolic and the number of eigenvalues with negative real parts are the same for A_{λ_0} and A_{λ_1} . Then, by Theorem 4.3.6 the flows $e^{A_{\lambda_0}t}$ and $e^{A_{\lambda_1}t}$ are topologically conjugate. By the Hartman-Grobman theorem (Theorem 9.6.2), the flow of $\dot{x} = f(x, \lambda_0)$ close to $\tilde{x}(\lambda_0)$ and the flow of $\dot{x} = f(x, \lambda_1)$ close to $\tilde{x}(\lambda_1)$ are locally topologically conjugate. Therefore, the stability of $\tilde{x}(\lambda)$ does not change for any $\lambda \in B_\delta(\tilde{\lambda})$.

This raises the question of what happens when $D_x f(\tilde{x}, \tilde{\lambda})$ is not invertible. The lack of invertibility is equivalent to the existence of a zero eigenvalue. It turns out that the topology of the local flow at $\tilde{x}$ may change under small perturbation of the parameter λ . In other words, the flows may not be locally topologically conjugate anymore. In case this happens, we say that a *bifurcation* occurs. In this section, we consider such bifurcations arising from the simplest possible setting, that is when $D_x f(\tilde{x}, \tilde{\lambda})$ has a unique zero eigenvalue and a one dimensional parameter space ($m = 1$). This leads to the following definitions.

Definition 4.6.1. Consider $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$. A *saddle-node* for f is a point $(\tilde{x}, \tilde{\lambda}) \in \mathbb{R}^n \times \mathbb{R}$ such that

- (i) $f(\tilde{x}, \tilde{\lambda}) = 0$, and
- (ii) 0 is an eigenvalue of $D_x f(\tilde{x}, \tilde{\lambda})$ with algebraic multiplicity one and all other eigenvalues have non-zero real parts.

Definition 4.6.2. Given $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ a *saddle-node bifurcation* occurs at the saddle-node $(\tilde{x}, \tilde{\lambda}) \in \mathbb{R}^n \times \mathbb{R}$ if the following conditions are met.

SNB1 There exists a smooth curve $g: (-\delta, \delta) \rightarrow \mathbb{R}^n \times \mathbb{R}$ denoted by $s \mapsto (g_1(s), g_2(s))$ such that $g(0) = (\tilde{x}, \tilde{\lambda})$ and $f(g_1(s), g_2(s)) = 0$.

SNB2 The curve defined by g has a quadratic tangency with $\mathbb{R}^n \times \{\tilde{\lambda}\}$ at $(\tilde{x}, \tilde{\lambda})$; that is,

$$g_2(0) = \tilde{\lambda}, \quad g_2'(0) = 0, \quad \text{and} \quad g_2''(0) \neq 0.$$

SNB3 If $s \neq 0$ then $D_x f(g_1(s), g_2(s))$ is hyperbolic and if $\sigma(s)$ is the eigenvalue of $D_x f(g_1(s), g_2(s))$ that satisfies $\sigma(0) = 0$, then $\sigma'(0) \neq 0$.

Before stating a general theorem we consider the simplest setting where the phase space is one dimensional.

Proposition 4.6.3. Consider a C^2 function $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} : (x, \lambda) \mapsto f(x, \lambda)$ satisfying

$$f(0, 0) = 0, \quad D_x f(0, 0) = 0, \quad D_x^2 f(0, 0) \neq 0, \quad \text{and} \quad D_\lambda f(0, 0) \neq 0.$$

Then, there is a saddle-node bifurcation at $(0, 0)$.

Proof. Since $D_\lambda f(0, 0) \neq 0$ the implicit function theorem implies the existence of a curve $g_2: (-\delta, \delta) \rightarrow \mathbb{R}$ such that $g_2(0) = 0$ and $f(s, g_2(s)) = 0$. Observe that by defining $g_1(s) = s$, **SNB1** is satisfied.

Differentiating $f(s, g_2(s)) = 0$ with respect to s gives

$$D_x f(s, g_2(s)) + D_\lambda f(s, g_2(s))g_2'(s) = 0 \tag{4.26}$$

and in particular for $s = 0$,

$$D_x f(0, 0) + D_\lambda f(0, 0)g_2'(0) = 0.$$

By assumption $D_x f(0, 0) = 0$ and $D_\lambda f(0, 0) \neq 0$, hence $g_2'(0) = 0$. Differentiating (4.26) gives

$$D_x^2 f(s, g_2(s)) + 2D_\lambda D_x f(s, g_2(s))g_2'(s) + D_\lambda^2 f(s, g_2(s))(g_2'(s))^2 + D_\lambda f(s, g_2(s))g_2''(s) = 0.$$

At the saddle-node we have

$$D_x^2 f(0, 0) + D_\lambda f(0, 0)g_2''(0) = 0$$

and hence

$$g_2''(0) = -\frac{D_x^2 f(0, 0)}{D_\lambda f(0, 0)} \neq 0.$$

This implies **SNB2**.

To check that **SNB3** is satisfied observe that

$$\frac{d}{ds} D_x f(s, g_2(s)) = D_x^2 f(s, g_2(s)) + D_\lambda D_x f(s, g_2(s))g_2'(s)$$

and hence at $s = 0$,

$$\frac{d}{ds} D_x f(0, 0) = D_x^2 f(0, 0) \neq 0$$

which implies that for $|\delta|$ sufficiently small $D_x f(s, g_2(s)) \neq 0$. $\square$

The following important result provides sufficient generic conditions in the setting of an n -dimensional phase space for the occurrence of a saddle-node bifurcation.

Theorem 4.6.4. *Assume $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is C^1 , $(\tilde{x}, \tilde{\lambda}) \in \mathbb{R}^n \times \mathbb{R}$ is a saddle-node, and the kernel of $D_x f(\tilde{x}, \tilde{\lambda})$ is spanned by the non-zero vector $\tilde{v} \in \mathbb{R}^n$. If*

$$u_1 \stackrel{\text{def}}{=} D_\lambda f(\tilde{x}, \tilde{\lambda}) \neq 0, \quad u_2 \stackrel{\text{def}}{=} D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \tilde{v}) \neq 0 \text{ and } u_1, u_2 \notin \text{image}\left(D_x f(\tilde{x}, \tilde{\lambda})\right), \quad (4.27)$$

then the conditions (SNB1), (SNB2) and (SNB3) are met and therefore there is a saddle-node bifurcation at $(\tilde{x}, \tilde{\lambda})$. Moreover, among all C^∞ one-parameter families that have a saddle-node, those that undergo a saddle-node bifurcation form an open and dense subset.

Proof. See the proof of Theorem 8.12 in [10]. $\square$

The following lemma provides a means of checking that a given vector is not in the range of matrix with a one-dimensional kernel. In practice, it will be useful to verify the last condition of (4.27).

Lemma 4.6.5. *Let $D: \mathbb{R}^n \rightarrow \mathbb{R}^n$ a linear transformation with $\dim(\ker(D)) = 1$. Then $\ker D^T = \langle w \rangle$, for some $w \in \mathbb{R}^n \setminus \{0\}$. Also, $u \in \text{image}(D)$ if and only if $u \cdot w = 0$.*

Proof. By the rank-nullity theorem, $\text{rank}(D) \stackrel{\text{def}}{=} \dim(\text{image}(D)) = n - \dim(\ker(D)) = n - 1$. Since $\text{rank}(D) = \text{rank}(D^T)$ the rank-nullity theorem implies that $\dim(\ker(D^T)) = 1$. Hence, there is a non-zero vector $w \in \mathbb{R}^n$ such that $\ker D^T = \langle w \rangle$.

Now, if $u \in \text{image}(D)$, there exists $y \in \mathbb{R}^n$ such that $u = Dy$, and then $u \cdot w = w^T u = w^T (Dy) = (w^T D)y = (D^T w)^T y = 0$. This implies that

$$\text{image}(D) \subset (\ker(D^T))^\perp = (\langle w \rangle)^\perp \stackrel{\text{def}}{=} \{u \in \mathbb{R}^n \mid u \cdot w = 0\}. \quad (4.28)$$

Conversely, assume $u \cdot w = 0$, that is $u \in (\langle w \rangle)^\perp = (\ker(D^T))^\perp$. Since $(\ker(D^T))^\perp$ is an $(n-1)$ -dimensional subspace and $\dim(\text{image}(D)) = n-1$, we use (4.28) to get $\text{image}(D) = (\ker(D^T))^\perp$. Hence, $u \in (\ker(D^T))^\perp = \text{image}(D)$, that is u is in the range of D . $\square$

As an application of Theorem 4.6.4 and Lemma 4.6.5, we now show the existence of a saddle-node bifurcation in the FitzHugh-Nagumo equation.

Example 4.6.6. Consider the FitzHugh-Nagumo equation

$$\dot{x} = f(x, \gamma) = \begin{pmatrix} x_1(x_1 - a)(1 - x_1) - x_2 \\ \epsilon(x_1 - \gamma x_2) \end{pmatrix}, \quad (4.29)$$

with parameter values $(a, \epsilon) = (5, 1)$ and where γ is left as a free parameter. In this case, the problem is simple enough so that we can do everything by hand and apply directly Theorem 4.6.4 to show the existence of a saddle-node bifurcation.

Numerically we identified a potential saddle-node point using the pseudo-arclength continuation algorithm presented in Section 3.1.2. See Figure 4.1 for a picture of a branch.

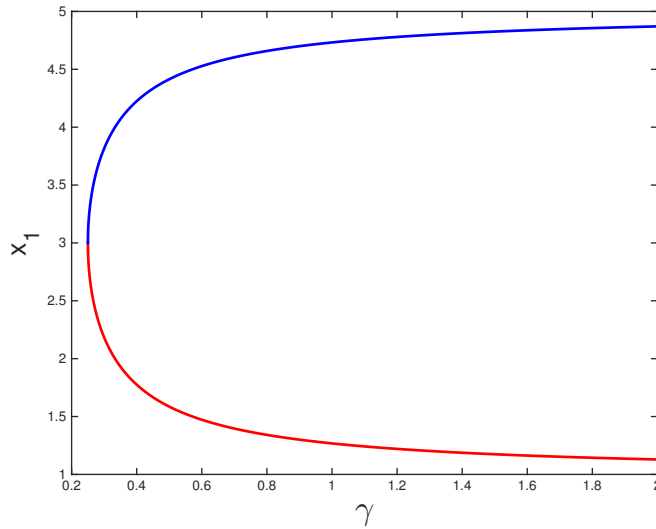


Figure 4.1: A branch of equilibrium solutions of the FitzHugh-Nagumo equation (4.29) computed using the pseudo-arclength continuation algorithm presented in Section 3.1.2. The computations were obtained by fixing $(a, \epsilon) = (5, 1)$ and leaving γ as a free parameter.

The computations suggest that there is a saddle-node bifurcation at $(\tilde{x}, \tilde{\gamma})$, with $\tilde{x} = (3, 12)$ and $\tilde{\gamma} = \frac{1}{4}$. Direct computation gives

$$f(\tilde{x}, \tilde{\gamma}) = \begin{pmatrix} 3(3-5)(1-3) - 12 \\ 1(3 - \frac{1}{4} \cdot 12) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \text{and} \quad D \stackrel{\text{def}}{=} D_x f(\tilde{x}, \tilde{\gamma}) = \begin{pmatrix} 4 & -1 \\ 1 & -\frac{1}{4} \end{pmatrix},$$

and the eigenvalues of D are given by 0 and $\frac{15}{4}$. Therefore, $(\tilde{x}, \tilde{\gamma})$ is a saddle-node.

Direct computations show that $\tilde{v} = (1, 4) \in \ker D$ and that

$$\begin{aligned} u_1 &\stackrel{\text{def}}{=} D_\gamma f(\tilde{x}, \tilde{\gamma}) = \begin{pmatrix} 0 \\ -\epsilon \tilde{x}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -12 \end{pmatrix} \neq 0 \\ u_2 &\stackrel{\text{def}}{=} D_x^2 f(\tilde{x}, \tilde{\gamma})(\tilde{v}, \tilde{v}) = \begin{pmatrix} 2\tilde{v}_1^2(a - 3\tilde{x}_1 + 1) \\ 0 \end{pmatrix} = \begin{pmatrix} 2(5 - 3 \cdot 3 + 1) \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \end{pmatrix} \neq 0. \end{aligned}$$

To show that $u_1, u_2 \notin \text{image}(D_x f(\tilde{x}, \tilde{\gamma}))$, we use Lemma 4.6.5. This requires computing $w \in \ker D^T$. Since $\dim \ker D^T = \dim \ker D = 1$, we can directly solve for the eigenpair $(0, w)$ for D^T to obtain $w = (1, -4) \neq 0$. In particular, $\ker D^T = \langle w \rangle$. Thus,

$$u_1 \cdot w = (0, -12)(1, -4)^T = 48 \neq 0 \quad \text{and} \quad u_2 \cdot w = (-6, 0)(1, -4)^T = -6 \neq 0.$$

By Lemma 4.6.5, $u_1, u_2 \notin \text{image}(D_x f(\tilde{x}, \tilde{\gamma}))$. Hence, all conditions in (4.27) are satisfied. By Theorem 4.6.4, a saddle-node bifurcation occurs in the FitzHugh-Nagumo equation at $(\tilde{x}, \tilde{\gamma}) = (3, 12) \in \mathbb{R}^2$.

Our success in directly applying Theorem 4.6.4 to demonstrate the existence of a saddle-node bifurcation for the FitzHugh-Nagumo equation made use of the fact that we had explicit values for the saddle-node $(\tilde{x}, \tilde{\gamma})$ and that we could perform all the linear algebra computations by hand. Following the general point of view of this book we are interested in recasting Theorem 4.6.4 in a way that we can rigorously verify that saddle-node bifurcations occur for general high-dimensional nonlinear ODEs. This is the purpose of the following theorem.

Theorem 4.6.7. *Given $X = (x, \lambda, v) \in \mathbb{R}^{2n+1}$ define the map $F : \mathbb{R}^{2n+1} \rightarrow \mathbb{R}^{2n+1}$ by*

$$F(X) \stackrel{\text{def}}{=} \begin{pmatrix} f(x, \lambda) \\ \ell(v) - 1 \\ D_x f(x, \lambda)v \end{pmatrix}, \quad (4.30)$$

where $\ell(v) = \xi \cdot v \in \mathbb{R}$ for a fixed vector $\xi \in \mathbb{R}^n \setminus \{0\}$, that is $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear. If $\tilde{X} = (\tilde{x}, \tilde{\lambda}, \tilde{v}) \in \mathbb{R}^{2n+1}$ is a nondegenerate zero of F and if $n - 1$ eigenvalues of $D_x f(\tilde{x}, \tilde{\lambda})$ have nonzero real parts, then a saddle-node bifurcation occurs at the saddle-node $(\tilde{x}, \tilde{\lambda})$.

Proof. Let $\tilde{X} = (\tilde{x}, \tilde{\lambda}, \tilde{v}) \in \mathbb{R}^{2n+1}$ such that $F(\tilde{X}) = 0$. Then $f(\tilde{x}, \tilde{\lambda}) = 0$ and $D_x f(\tilde{x}, \tilde{\lambda})\tilde{v} = 0$, which implies that $0 \in \sigma(D_x f(\tilde{x}, \tilde{\lambda}))$. The rest of the proof of the theorem follows the same idea over and over again: prove each statement (i)-(v) below by contradiction by producing a vector $X \in \mathbb{R}^{2n+1} \setminus \{0\}$ such that $D_X F(\tilde{X})X = 0$, violating the nondegeneracy of $\tilde{X}$. The statements we will prove using that strategy are the following.

- (i) 0 has geometric multiplicity one;

- (ii) $u_2 \stackrel{\text{def}}{=} D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \tilde{v}) \neq 0$;
- (iii) $u_2 \notin \text{image}(D_x f(\tilde{x}, \tilde{\lambda}))$;
- (iv) $u_1 \stackrel{\text{def}}{=} D_\lambda f(\tilde{x}, \tilde{\lambda}) \neq 0$;
- (v) $u_1 \notin \text{image}(D_x f(\tilde{x}, \tilde{\lambda}))$.

Since the phase condition $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear, there exists $\xi \in \mathbb{R}^n \setminus \{0\}$ such that $\ell(v) = \xi \cdot v$. Then the Jacobian matrix of F at $\tilde{X}$ is given by

$$D_X F(\tilde{X}) = \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix}. \quad (4.31)$$

(i) We show that 0 is an eigenvalue with geometric multiplicity one. By contradiction, assume the opposite, that is $\dim(\ker(D_x f(\tilde{x}, \tilde{\lambda}))) > 1$. In this case, let $\tilde{u} \in \mathbb{R}^n$ linearly independent from $\tilde{v}$ such that $D_x f(\tilde{x}, \tilde{\lambda})\tilde{u} = 0$. For all $\alpha \in \mathbb{R}$,

$$v_\alpha \stackrel{\text{def}}{=} (1 - \alpha \ell(\tilde{u})) \tilde{v} + \alpha \tilde{u}$$

satisfies $\ell(v_\alpha) = (1 - \alpha \ell(\tilde{u}))\ell(\tilde{v}) + \alpha \ell(\tilde{u}) = (1 - \alpha \ell(\tilde{u})) + \alpha \ell(\tilde{u}) = 1$, for all $\alpha \in \mathbb{R}$. This in turn implies that for all scalars $\alpha \in \mathbb{R}$, we have

$$F(\tilde{x}, \tilde{\lambda}, v_\alpha) = \begin{pmatrix} f(\tilde{x}, \tilde{\lambda}) \\ \ell(v_\alpha) - 1 \\ D_x f(\tilde{x}, \tilde{\lambda})v_\alpha \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

which implies that the zero $\tilde{X} = (\tilde{x}, \tilde{\lambda}, \tilde{v})$ is not isolated, and therefore $D_X F(\tilde{X}) \in M_{2n+1}(\mathbb{R})$ is not invertible. In fact, $X_1 \stackrel{\text{def}}{=} (0, 0, -\ell(\tilde{u})\tilde{v} + \tilde{u}) \neq 0$ satisfies

$$\begin{aligned} D_X F(\tilde{X})X_1 &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -\ell(\tilde{u})\tilde{v} + \tilde{u} \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ -\ell(\tilde{u})\xi \cdot \tilde{v} + \xi \cdot \tilde{u} \\ -\ell(\tilde{u})D_x f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x f(\tilde{x}, \tilde{\lambda})\tilde{u} \end{pmatrix} = \begin{pmatrix} 0 \\ -\ell(\tilde{u})\ell(\tilde{v}) + \ell(\tilde{u}) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

since $\ell(\tilde{v}) = 1$. This contradicts the fact that $\tilde{X}$ is a nondegenerate zero of F . Hence, $\dim(\ker(D_x f(\tilde{x}, \tilde{\lambda}))) = 1$ and 0 is an eigenvalue of $D_x f(\tilde{x}, \tilde{\lambda})$ with geometric multiplicity one. According to Definition 4.6.1, the point $(\tilde{x}, \tilde{\lambda}) \in \mathbb{R}^n \times \mathbb{R}$ is a saddle-node for f .

(ii) Assume that $u_2 = D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \tilde{v}) = 0$. Then $X_2 \stackrel{\text{def}}{=} (\tilde{v}, 0, 0) \neq 0$ satisfies

$$\begin{aligned} D_X F(\tilde{X})X_2 &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix} \begin{pmatrix} \tilde{v} \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda})\tilde{v} \\ 0 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

which once again violates the invertibility of $D_X F(\tilde{X})$. Therefore $u_2 \neq 0$.

(iii) Assume that $u_2 \in \text{image}(D_x f(\tilde{x}, \tilde{\lambda}))$. Then there exists $\hat{v} \notin \ker(D_x f(\tilde{x}, \tilde{\lambda}))$ such that $D_x f(\tilde{x}, \tilde{\lambda})\hat{v} = -u_2 \neq 0$. In this case, $X_3 \stackrel{\text{def}}{=} (\tilde{v}, 0, \hat{v} - \ell(\hat{v})\tilde{v}) \neq 0$ satisfies

$$\begin{aligned} D_X F(\tilde{X})X_3 &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix} \begin{pmatrix} \tilde{v} \\ 0 \\ \hat{v} - \ell(\hat{v})\tilde{v} \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ \ell(\hat{v}) - \ell(\hat{v})\ell(\tilde{v}) \\ D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \tilde{v}) + D_x f(\tilde{x}, \tilde{\lambda})(\hat{v} - \ell(\hat{v})\tilde{v}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ u_2 - u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

which once again violates the invertibility of $D_X F(\tilde{X})$. Hence $u_2 \notin \text{image}(D_x f(\tilde{x}, \tilde{\lambda}))$.

(iv) Assume that $u_1 = D_\lambda f(\tilde{x}, \tilde{\lambda}) = 0$. By Lemma 4.6.5, since $D \stackrel{\text{def}}{=} D_x f(\tilde{x}, \tilde{\lambda})$ is a linear transformation with $\dim(\ker(D)) = 1$, then $\ker D^T = \langle w \rangle$, for some $w \in \mathbb{R}^n \setminus \{0\}$. Now, $w \cdot u_2 \neq 0$, because otherwise, Lemma 4.6.5 would imply that $u_2 \in \text{image}(D)$, a contradiction with (iii). Since $w \cdot u_2 \neq 0$, we can determine uniquely $\hat{\alpha} \in \mathbb{R}$ a solution to the scalar linear equation

$$w \cdot (D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + \hat{\alpha}u_2) = 0.$$

By Lemma 4.6.5, $D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + \hat{\alpha}u_2 \in \text{image}(D_x f(\tilde{x}, \tilde{\lambda}))$ and so there exists $\hat{v} \in \mathbb{R}^n$ such that

$$D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + \hat{\alpha}u_2 + D_x f(\tilde{x}, \tilde{\lambda})\hat{v} = 0.$$

Hence, $X_4 \stackrel{\text{def}}{=} (\hat{\alpha}\tilde{v}, 1, \hat{v} - \ell(\hat{v})\tilde{v}) \neq 0$ satisfies

$$\begin{aligned} D_X F(\tilde{X})X_4 &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & 0 & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix} \begin{pmatrix} \hat{\alpha}\tilde{v} \\ 1 \\ \hat{v} - \ell(\hat{v})\tilde{v} \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ \ell(\hat{v}) - \ell(\hat{v})\ell(\tilde{v}) \\ \hat{\alpha}u_2 + D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x f(\tilde{x}, \tilde{\lambda})\hat{v} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

a contradiction, and therefore $u_1 = D_\lambda f(\tilde{x}, \tilde{\lambda}) \neq 0$.

(v) Finally, assume that $u_1 \in \text{image} \left(D_x f(\tilde{x}, \tilde{\lambda}) \right)$. Then there exists $\hat{x} \notin \ker \left(D_x f(\tilde{x}, \tilde{\lambda}) \right)$ such that $D_x f(\tilde{x}, \tilde{\lambda})\hat{x} = -u_1 = -D_\lambda f(\tilde{x}, \tilde{\lambda})$. Thus, for all $\alpha \in \mathbb{R}$,

$$D_\lambda f(\tilde{x}, \tilde{\lambda}) + D_x f(\tilde{x}, \tilde{\lambda})(\hat{x} + \alpha \tilde{v}) = 0 \in \mathbb{R}^n. \quad (4.32)$$

Now, since

$$D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x} + \alpha \tilde{v}) = D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x}) + \alpha u_2,$$

and since $w \cdot u_2 \neq 0$ (recall that this follows from (iii) and Lemma 4.6.5), we can determine uniquely a value $\hat{\alpha} \in \mathbb{R}$ such that

$$w \cdot \left(D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x}) + \hat{\alpha} u_2 \right) = 0.$$

By Lemma 4.6.5, $D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x}) + \hat{\alpha} u_2 \in \text{image} \left(D_x f(\tilde{x}, \tilde{\lambda}) \right)$, and so we can find $\hat{v} \in \mathbb{R}^n$ such that

$$D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x}) + \hat{\alpha} u_2 + D_x f(\tilde{x}, \tilde{\lambda})\hat{v} = 0.$$

This implies that $X_5 \stackrel{\text{def}}{=} (\hat{x} + \hat{\alpha} \tilde{v}, 1, \hat{v} - \ell(\hat{v})\tilde{v}) \neq 0$ satisfies

$$\begin{aligned} D_X F(\tilde{X})X_5 &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ 0 & 0 & \xi^T \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} & D_x f(\tilde{x}, \tilde{\lambda}) \end{pmatrix} \begin{pmatrix} \hat{x} + \hat{\alpha} \tilde{v} \\ 1 \\ \hat{v} - \ell(\hat{v})\tilde{v} \end{pmatrix} \\ &= \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda})(\hat{x} + \hat{\alpha} \tilde{v}) + D_\lambda f(\tilde{x}, \tilde{\lambda}) \\ 0 \\ D_x^2 f(\tilde{x}, \tilde{\lambda})(\tilde{v}, \hat{x}) + \hat{\alpha} u_2 + D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v} + D_x f(\tilde{x}, \tilde{\lambda})\hat{v} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

where we applied (4.32) to get that the first entry of $D_X F(\tilde{X})X_5$ is $0 \in \mathbb{R}^n$. This is a contradiction, and therefore $u_1 \notin \text{image} \left(D_x f(\tilde{x}, \tilde{\lambda}) \right)$.

Combining (i), (ii), (iii), (iv) and (v), we have shown that the conditions (4.27) hold. By Theorem 4.6.4, we conclude that there is a saddle-node bifurcation at $(\tilde{x}, \tilde{\lambda})$. $\square$

One of the assumption of Theorem 4.6.7 is the nondegeneracy of the zero $\tilde{X}$ of F . As is made clear in the conclusion of Theorem 2.4.2, if we use the radii polynomial approach to obtain the existence of $\tilde{X}$, then $D_X F(\tilde{X})$ is automatically invertible, and the nondegeneracy of $\tilde{X}$ comes *for free*.

The general strategy to prove existence of saddle-node bifurcation is now clear. First, we use the radii polynomial approach to prove the existence of a zero $\tilde{X} \in \mathbb{R}^{2n+1}$ of $F(X) = 0$ where F is defined in (4.30). By Theorem 2.4.2, $\tilde{X}$ is nondegenerate. Then,

using the theory introduced in Section 4.5, we show that all the non zero eigenvalues of $D_x f(\tilde{x}, \tilde{\lambda})$ have nonzero real parts. More explicitly, having $n - 1$ numerical approximations $\{(\tilde{\lambda}_j, \tilde{v}^j) \mid j = 2, \dots, n\} \subset \mathbb{C}$ such that $D_x f(\tilde{x}, \tilde{\lambda})\tilde{v}^j \approx \tilde{\lambda}_j \tilde{v}^j$, compute eigenpairs $\{(\tilde{\lambda}_j, \tilde{v}^j) \mid j = 2, \dots, n\} \subset \mathbb{C}$ such that $(\tilde{\lambda}_j, \tilde{v}^j) \in \overline{B_{r_j}(\tilde{\lambda}_j, \tilde{v}^j)}$ for some radii r_j , $j = 2, \dots, n$ and then verify that

$$\overline{B_{r_j}(\tilde{\lambda}_j)} \cap i\mathbb{R} = \emptyset, \quad \text{for } j = 2, \dots, n. \quad (4.33)$$

If (4.33) can be verified, then all hypotheses of Theorem 4.6.7 are satisfied and we can conclude that a saddle-node bifurcation occurs at $(\tilde{x}, \tilde{\lambda})$.

We now apply the proposed approach to prove that two saddle-node bifurcations occur in the Hodgkin-Huxley model.

Example 4.6.8. The Hodgkin-Huxley model for the action potential of a space-clamped squid axon is defined by the four dimensional vector field

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \begin{pmatrix} -20 - 120x_2^3x_3 \left(x_1 - 25.1 \ln\left(\frac{23}{1350}(550 - \lambda)\right) \right) - 36x_4^4 \left(x_1 - 25.1 \ln\left(\frac{\lambda}{400}\right) \right) - 0.3(x_1 + 24.3) \\ \frac{9}{25}(1 - x_2) \frac{x_1 - \Delta(\lambda) + 35}{1 - \exp\left(-\frac{x_1 - \Delta(\lambda) + 35}{10}\right)} - \frac{72}{5}x_2 \exp\left(-\frac{x_1 - \Delta(\lambda) + 60}{18}\right) \\ \frac{63}{250}(1 - x_3) \exp\left(-\frac{x_1 - \Delta(\lambda) + 60}{20}\right) - \frac{18}{5}x_3 \frac{1}{\exp\left(-\frac{x_1 - \Delta(\lambda) + 30}{10}\right) + 1} \\ \frac{9}{250}(1 - x_4) \frac{x_1 - \Delta(\lambda) + 50}{1 - \exp\left(-\frac{x_1 - \Delta(\lambda) + 50}{10}\right)} - \frac{9}{20}x_4 \exp\left(-\frac{x_1 - \Delta(\lambda) + 60}{80}\right) \end{pmatrix} \quad (4.34)$$

where

$$\Delta(\lambda) \stackrel{\text{def}}{=} 9.32 \ln\left(\frac{11}{10} - \frac{\lambda}{500}\right).$$

The variable x_1 is the membrane potential, x_2 is the activation of a sodium current, x_3 is the activation of a potassium current, x_4 is the inactivation of the sodium current and the parameter λ is the external potassium concentration.

Using the pseudo-arclength continuation technique of Section 3.1.2, we obtain the bifurcation diagram of Figure 4.2. From this numerical simulation, we can conjecture the existence of two saddle-node bifurcations at $\lambda \approx 426.42$ and at $\lambda \approx 53.61$.

Denote by $f(x, \lambda)$ the right-hand side of (4.34). Let $X = (x, \lambda, v) \in \mathbb{R}^9$, and define $F : \mathbb{R}^9 \rightarrow \mathbb{R}^9$ as in (4.30). Applying Newton's method to problem (4.30), we compute two

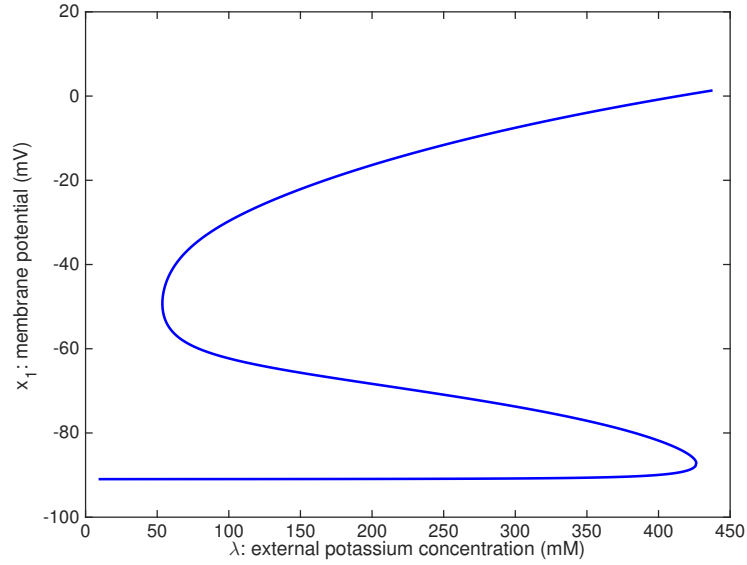


Figure 4.2: A branch of equilibria which undergoes two saddle-node bifurcations.

approximate solutions

$$\bar{X}_1 \stackrel{\text{def}}{=} \begin{pmatrix} -87.2515605439908 \\ 0.0089034888748 \\ 0.9231857631959 \\ 0.1363687823857 \\ 426.4159725555050 \\ 0.9998982043007 \\ 0.0011653907919 \\ -0.0105518817291 \\ 0.0095331365494 \end{pmatrix} \quad \text{and} \quad \bar{X}_2 \stackrel{\text{def}}{=} \begin{pmatrix} -49.270516282150226 \\ 0.170894916154782 \\ 0.241946685250340 \\ 0.487852905937695 \\ 53.607108413683697 \\ 0.999427022035232 \\ 0.016569867126847 \\ -0.025152384230317 \\ 0.015441006985167 \end{pmatrix} \quad (4.35)$$

corresponding to the two possible saddle-node points.

Theorem 4.6.9. *There is a saddle-node bifurcation at a point*

$$(\tilde{x}_1, \tilde{\lambda}_1) \in \overline{B_{r_1}((\bar{x}_1, \bar{\lambda}_1))}, \quad r_1 \stackrel{\text{def}}{=} 1.715 \times 10^{-12},$$

where $(\bar{x}_1, \bar{\lambda}_1) \in \mathbb{R}^5$ is given by the first five components of $\bar{X}_1$ in (4.35).

Theorem 4.6.10. *There is a saddle-node bifurcation at a point*

$$(\tilde{x}_2, \tilde{\lambda}_2) \in \overline{B_{r_2}((\bar{x}_2, \bar{\lambda}_2))}, \quad r_2 \stackrel{\text{def}}{=} 2.034 \times 10^{-12},$$

where $(\bar{x}_2, \bar{\lambda}_2) \in \mathbb{R}^5$ is given by the first five components of $\bar{X}_2$ in (4.35).

Proof. The proof has two parts. In part (a) we use the radii polynomial approach to prove existence of two nondegenerate zeros $\tilde{X}_1 = (\tilde{x}_1, \tilde{\lambda}_1, \tilde{v}_1) \in \mathbb{R}^9$ and $\tilde{X}_2 = (\tilde{x}_2, \tilde{\lambda}_2, \tilde{v}_2) \in \mathbb{R}^9$ of the map F given in (4.30). In part (b) we show that all the nonzero eigenvalues of $D_x f(\tilde{x}_1, \tilde{\lambda}_1)$ and $D_x f(\tilde{x}_2, \tilde{\lambda}_2)$ have nonzero real parts. The proofs of Theorem 4.6.9 and Theorem 4.6.10 then follows from Theorem 4.6.7.

(a) Let $\bar{X}$ one of the two points $\tilde{X}_1$ or $\tilde{X}_2$ given in (4.35). We already have that $F(\bar{X}) \approx 0$, with F given by (4.30). Note that

$$D_X F(\bar{X}) = \begin{pmatrix} D_x f(\bar{x}, \bar{\lambda}) & D_\lambda f(\bar{x}, \bar{\lambda}) & 0 \\ 0 & 0 & 2\bar{v}^T \\ D_x (D_x f(\bar{x}, \bar{\lambda})\bar{v}) & D_{x,\lambda} f(\bar{x}, \bar{\lambda})\bar{v} & D_x f(\bar{x}, \bar{\lambda}) \end{pmatrix}. \quad (4.36)$$

Let $A = D_X F(\bar{X})^{-1}$. Apply the localized radii polynomial approach (Theorem 2.5.2) with $r^* = 10^{-4}$ to get the existence intervals

$$I_1 = [1.715 \times 10^{-12}, 8.052 \times 10^{-6}] \quad \text{and} \quad I_2 = [2.034 \times 10^{-12}, 1.974 \times 10^{-5}],$$

respectively. Since $8.052 \times 10^{-6}, 1.974 \times 10^{-5} < r^* = 10^{-4}$, then the existence intervals are valid. Let $r_1 \stackrel{\text{def}}{=} 1.715 \times 10^{-12}$ and $r_2 \stackrel{\text{def}}{=} 2.034 \times 10^{-12}$. Recall (4.35), then by Theorem 2.5.2, there exists a unique $\tilde{X}_1 = (\tilde{x}_1, \tilde{\lambda}_1, \tilde{v}_1) \in \overline{B_{r_1}(\bar{X}_1)}$ such that $F(\tilde{X}_1) = 0$ and there exists a unique $\tilde{X}_2 = (\tilde{x}_2, \tilde{\lambda}_2, \tilde{v}_2) \in \overline{B_{r_2}(\bar{X}_2)}$ such that $F(\tilde{X}_2) = 0$. Hence, for $j = 1, 2$, $f(\tilde{x}_j, \tilde{\lambda}_j) = 0$ and the kernel of $D_x f(\tilde{x}_j, \tilde{\lambda}_j)$ must be one dimensional, as otherwise we would not have that $\tilde{X}_j$ isolated solution in $\mathbb{R}^9$.

(b) Choose $j \in \{1, 2\}$ and let $I \stackrel{\text{def}}{=} I_j$ the existence interval associated to $\bar{X} \stackrel{\text{def}}{=} \bar{X}_j$. Let r the smallest radius of the existence interval I . Define $\mathbf{B} = \overline{B_r((\bar{x}, \bar{\lambda}))} \subset \mathbb{R}^5$, that is

$$\mathbf{B} = \prod_{k=1}^4 [\bar{x}_k - r, \bar{x}_k + r] \times [\bar{\lambda} - r, \bar{\lambda} + r].$$

Let $D \stackrel{\text{def}}{=} D_x f(\tilde{x}, \tilde{\lambda})$ and $\mathbf{D} \stackrel{\text{def}}{=} D_x f(\mathbf{B})$ a 4×4 interval matrix computed with interval arithmetic. Note that $D \subset \mathbf{D}$. Using the radii polynomial approach as introduced in Section 4.5, we show that $\sigma(\mathbf{D}) \subset \bigcup_{j=1}^n B_j$, for some small balls $B_j \in \mathbb{C}$. We proved that the eigenvalues of $D_x f(\tilde{x}_1, \tilde{\lambda}_1)$ are enclosed in $\bigcup_{j=1}^4 B_j$, where

$$\begin{aligned} B_1 &= \{z \in \mathbb{C} : |z + 32.02633660454969| \leq 3.394274197807681 \times 10^{-11}\} \\ B_2 &= \{z \in \mathbb{C} : |z - 9.305978062941147 \times 10^{-16}| \leq 1.069542699059249 \times 10^{-11}\} \\ B_3 &= \{z \in \mathbb{C} : |z + 0.9317141708275124| \leq 9.975910428070430 \times 10^{-12}\} \\ B_4 &= \{z \in \mathbb{C} : |z + 0.5558951614569074| \leq 1.774408327300926 \times 10^{-12}\}. \end{aligned}$$

Therefore, we obtain that $0 \in B_k$ for the unique $k = 2 \in \{1, 2, 3, 4\}$, and that $B_j \cap i\mathbb{R} = \emptyset$, for all $j \in \{1, 3, 4\}$. This shows that $(\tilde{x}_1, \tilde{\lambda}_1)$ is a saddle-node. We repeated the same procedure to show that $(\tilde{x}_2, \tilde{\lambda}_2)$ is also a saddle-node. $\square$

4.7 Exercises

Exercise 4.7.1. Prove Proposition 4.1.5.

Hints: (i) Observe that $(BCB^{-1})^m = BC^m B^{-1}$.

(ii) Recall the binomial theorem

$$(A + B)^m = m! \sum_{i+j=m} \frac{1}{i!j!} A^i B^j.$$

Exercise 4.7.2. Prove that if $M(v_1(t), v_2(t), \dots, v_n(t))$ is a multilinear function, then

$$\frac{dM}{dt}(t) = \sum_{k=1}^n M\left(v_1(t), \dots, v_{k-1}(t), \frac{dv_k}{dt}(t), v_{k+1}, \dots, v_n(t)\right).$$

Exercise 4.7.3. Prove Proposition 4.2.7.

Exercise 4.7.4. Consider $\dot{x} = Ax$ and assume the real parts of all eigenvalues of A are negative.

(i) Prove that $\{x \mid \|x\| \leq 1\}$ is an attracting neighborhood for 0.

(ii) Prove that there exists a trapping region containing 0.

Exercise 4.7.5. Complete the proof of Theorem 4.3.6(i) and show that the conjugacy $h: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a homeomorphism. Hint: you can write down the inverse explicitly.

Exercise 4.7.6. Let X be a topological space. A set $G \subset X$ is *generic* if G is open and dense in X . Prove that hyperbolic matrices are open and dense in the space of linear maps on $\mathbb{R}^n$.

Hint: Given $A \in M_n(\mathbb{R})$, show that $\|A\| := \max_{1 \leq i, j \leq n} |a_{i,j}|$ defines a metric on $M_n(\mathbb{R})$.

Exercise 4.7.7. Prove part (i) of Theorem 4.3.5. (Hint: use (4.4).)

Exercise 4.7.8. Let $M \in M_n(\mathbb{R})$ and consider the problem:

$$g(\lambda, v) = \begin{pmatrix} (M - \lambda I)v \\ \|v\|^2 - 1 \end{pmatrix} = 0.$$

Show how to compute a real eigenpair for M by solving the above problem using the radii polynomial approach.

Exercise 4.7.9. Add an exercise on how to rigorously compute generalized eigenvectors.

Exercise 4.7.10. Compute the eigenpairs of the Jacobian matrix of an equilibrium by defining a interval matrix containing the true Jacobian (using the error bounds from the radii polynomials).

Exercise 4.7.11. Prove the existence of two saddle-node bifurcations in the model (3.25) at the fixed parameter values $\alpha = 0.25$, $\beta = 4$ and $\gamma = 0.5$, with λ the free parameter. You may use Figure 3.9 as a starting point.

Exercise 4.7.12. Prove that there are saddle-node bifurcations $(\tilde{x}_1, \lambda_1)$ and $(\tilde{x}_2, \lambda_2)$ with $0 < \lambda_1 < \lambda_2 < 1$ for the map $f : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^2$ given by (3.27).

Exercise 4.7.13 (Existence and uniqueness for linear differential equations). Consider $J \subset \mathbb{R}$ an open interval. Suppose that $g : J \rightarrow \mathbb{R}^n$ and $a_{i,j} : J \rightarrow \mathbb{R}$, $1 \leq i, j \leq n$, are continuous functions. Set

$$A(t) = \begin{pmatrix} a_{1,1}(t) & \cdots & a_{1,n}(t) \\ \vdots & \ddots & \vdots \\ a_{n,1}(t) & \cdots & a_{n,n}(t) \end{pmatrix}.$$

Prove that for any $t_0 \in J$ the initial value problem

$$\dot{x}(t) = A(t)x + g(t), \quad x(t_0) = x_0, \quad (4.37)$$

has a unique solution $x : J \rightarrow \mathbb{R}^n$.

Hint: Define an *integrating factor* $C(t)$ by writing down a solution of the equation

$$C'(t) = -C(t)A(t),$$

that is write down an explicit formula for such a $C(t)$. Show directly that

- $C(t)$ is well defined, continuous, and differentiable on J .
- $C(t_0) = Id$.
- $C(t)$ is invertible for every $t \in J$.

Now multiply both sides of Equation (4.37) by $C(t)$, apply the product rule on the left, integrate both sides from t_0 to t with $t \in J$, and solve for $x(t)$. This provides an explicit formula for the solution $x(t)$. Show that the formula gives a well defined, continuous, differentiable function for each $t \in J$.

Chapter 5

Elementary Dynamics of Differential Equations

The emphasis of the previous chapter was on linear systems wherein the concept of a flow was introduced. In this chapter we discuss simple properties of flows with a focus on asymptotic behavior. This leads us to a more geometric perspective.

5.1 Invariant Dynamics of Flows

With the language of flows we can generalize the concept of an orbit.

Definition 5.1.1. A set $S \subset X$ is an *invariant set* for the flow $\varphi: \mathbb{R} \times X \rightarrow X$ if

$$\varphi(\mathbb{R}, S) = S.$$

Observe that every orbit is an invariant set and any invariant set is the union of orbits. Furthermore, if S is an invariant set for φ then $\varphi(t, S) = S$ for all $t \in \mathbb{R}$.

Slightly weaker, but useful notions of invariance include the following.

Definition 5.1.2. A set $S \subset X$ is *forward invariant* if $\varphi([0, \infty), S) \subset S$ and is *backward invariant* if $\varphi((-\infty, 0], S) \subset S$.

Observe that an invariant set is both forward and backward invariant.

Definition 5.1.3. A point $x \in X$ is a *equilibrium point* if $\varphi(\mathbb{R}, x) = x$ or equivalently if $\varphi(t, x) = x$ for all $t \in \mathbb{R}$.

Observe that if the flow φ is generated by an ODE $\dot{x} = f(x)$, then x_0 is an equilibrium point if and only if $f(x_0) = 0$.

Example 5.1.4. The *logistic equation* is used in biology as a simple model for population growth that includes overcrowding effects. It takes the form

$$\dot{x} = rx(K - x) \quad (5.1)$$

where $r > 0$ represents the birth rate and $K > 0$ is the carrying capacity for the environment. Explicit solutions to this equation take the form

$$\varphi(t, x_0) = \frac{x_0 K e^{rKt}}{K - x_0 + x_0 e^{rKt}}. \quad (5.2)$$

Observe that φ is not a flow on the phase space $\mathbb{R}$. If $x_0 < 0$ or $x_0 > K$, then $|\varphi(t, x_0)| \rightarrow \infty$ as $t \rightarrow \frac{1}{rK} \ln \left(\frac{x_0 - K}{x_0} \right)$.

The language of flows is useful. As is discussed in Section 4.2 it allows us to compare dynamics between different systems. There are several ways to resolve the challenge when an ODE does not generate a flow. The first is to insist that because K is the maximal population level that can be supported the population levels of interest lie in the interval $[0, K]$. Observe that the restriction $\varphi: \mathbb{R} \times [0, K] \rightarrow [0, K]$ is a flow. However, one can easily imagine scenarios in which populations, at least initially, exceed the carrying capacity, e.g., over grazing caused by the introduction of domestic animals. An alternative method is to make the scientific observation that populations levels are always non-negative and therefore we should consider the dynamics on $[0, \infty)$. This still does not make φ a flow, but observe that in forward time any solution to the differential equation limits to an element of either 0 or K . Thus given any biologically realistic initial condition x_0 , after a reasonable amount of time τ , $\varphi(\tau, x_0)$ will be close to $[0, K]$. As is describe in below this gives rise to the notion of an attractor. Yet another alternative, purely mathematical in nature, is to prove that an appropriate rescaling of the nonlinearity of the differential equation produces a flow that preserves the geometry of trajectories of the original ODE.

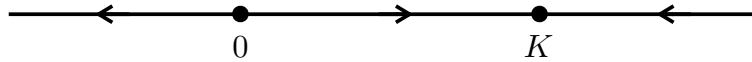


Figure 5.1: The phase portrait of the logistic equation (5.1) for $r, K > 0$.

The dynamics of the logistic equation restricted to $[0, K]$ is particularly simple (e.g. see Figure 5.1). There are three orbits: two equilibrium points and the orbit $(0, K)$ along which the dynamics moves from 0 to K . This suggests that it is worth naming the latter type of orbit.

Definition 5.1.5. A point $x_0 \in X$ is a *heteroclinic point* of a flow $\varphi: \mathbb{R} \times X \rightarrow X$ if

$$\lim_{t \rightarrow \infty} \varphi(t, x_0) = x_+ \quad \text{and} \quad \lim_{t \rightarrow -\infty} \varphi(t, x_0) = x_-$$

where $x_- \neq x_+$ are equilibria. The orbit $\varphi(\mathbb{R}, x_0)$ is called a *heteroclinic orbit* from x_- to x_+ . In case $x_- = x_+$ we say that x_0 is a *homoclinic point* and $\varphi(\mathbb{R}, x_0)$ is called a *homoclinic orbit*.

We leave the proof of the following proposition as an exercise.

Proposition 5.1.6. *If $f: \mathbb{R} \rightarrow \mathbb{R}$ is locally Lipschitz continuous, then every bounded solution to*

$$\dot{x} = f(x)$$

is either an equilibrium or a heteroclinic orbit.

Definition 5.1.7. A point $x \in X$ is a *periodic point* if there exists $T > 0$ such that $\varphi(T, x) = x$. The associated *periodic orbit* is $\varphi([0, T], x)$.

Observe that a periodic orbit is an invariant set, since by Definition 4.2.1, $\varphi(\mathbb{R}, x) = \varphi([0, T], x)$.

Example 5.1.8. Consider the system of differential equations

$$\begin{aligned}\dot{x}_1 &= -x_2 + \lambda x_1(K^2 - x_1^2 - x_2^2) \\ \dot{x}_2 &= x_1 + \lambda x_2(K^2 - x_1^2 - x_2^2).\end{aligned}$$

Changing to polar coordinates results in the system

$$\begin{aligned}\dot{\theta} &= 1 \\ \dot{r} &= \lambda r(K^2 - r^2).\end{aligned}$$

Observe that the circle $r = |K|$ or equivalently $\Gamma \stackrel{\text{def}}{=} \{x \in \mathbb{R}^2 \mid \|x\| = |K|\}$ is a periodic orbit. Since the equation in r is a scalar differential equation, by Proposition 5.1.6 if $r_0 \in (0, K)$, then r_0 is a heteroclinic point. The associated heteroclinic orbit goes from 0 to K and thus solutions spiral away from the origin toward the periodic orbit.

Example 5.1.8 suggests that we want to be able to discuss limits of trajectories that are not single points. We begin with a general definition.

Definition 5.1.9. Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow and let $U \subset X$. The *alpha* and *omega limit sets* of U are defined by

$$\alpha(U) = \alpha(U, \varphi) \stackrel{\text{def}}{=} \bigcap_{T \leq 0} \overline{\varphi((-\infty, T], U)} \quad \text{and} \quad \omega(U) = \omega(U, \varphi) \stackrel{\text{def}}{=} \bigcap_{T \geq 0} \overline{\varphi([T, \infty), U)},$$

respectively.

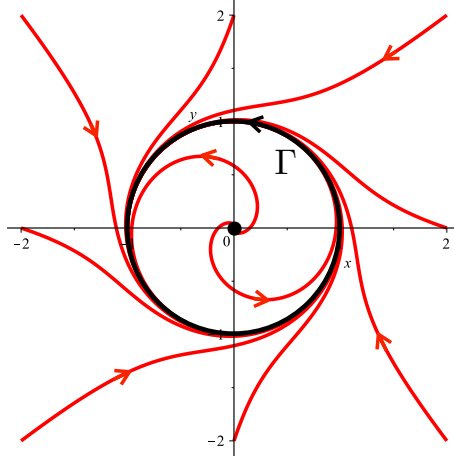


Figure 5.2: The phase portrait of the equation of Example 5.1.8 with $K = \lambda = 1$.

In a slight abuse of notation, if $x \in X$ then we will let $\omega(x) = \omega(\{x\})$ and $\alpha(x) = \alpha(\{x\})$. As an exercise we let the reader check that if x_0 is a heteroclinic point from x_- to x_+ , then $\alpha(x_0) = x_-$ and $\omega(x_0) = x_+$.

Going back to Example 5.1.8 with $K = \lambda = 1$, consider any $x_0 \neq 0$ with $\|x_0\|_2 < 1$. Then, one can verify that $\alpha(x_0) = \{0\}$ and $\omega(x_0) = \Gamma$. Moreover, if $\|x_0\|_2 > 1$, then $\alpha(x_0) = \emptyset$ and $\omega(x_0) = \Gamma$.

Proposition 5.1.10. *Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow and $U, V \subset X$. We have:*

(i) *If there exists $t \in \mathbb{R}$ such that $\varphi(t, U) = V$, then*

$$\omega(U) = \omega(V) \quad \text{and} \quad \alpha(U) = \alpha(V).$$

(ii) *A point $y \in \omega(U, \varphi)$ if and only if there exists a sequence of times $t_k \rightarrow \infty$ and a sequence of points $x_k \in U$ such that*

$$\lim_{k \rightarrow \infty} \varphi(t_k, x_k) = y.$$

Furthermore, $\omega(U) = \omega(\varphi([0, \infty), U))$.

(iii) *Let V be a closed, forward invariant set. If $U \subset V$, then $\omega(U) \subset V$.*

(iv) *If U is forward invariant, then*

$$\omega(U) = \bigcap_{t \geq 0} \overline{\varphi(t, U)}. \tag{5.3}$$

(v) $\omega(U)$ and $\alpha(U)$ are closed invariant sets. Furthermore, if U is a closed invariant set, then

$$U = \omega(U) = \alpha(U).$$

(vi) If $\varphi([0, \infty), U) \subset K$ where $K \subset X$ is compact, then $\omega(U)$ is nonempty and compact. If in addition, U is connected, then $\omega(U)$ is connected.

(vii) If $V \subset \omega(U)$, then $\omega(V) \subset \omega(U)$ and $\alpha(V) \subset \omega(U)$.

Similar results apply to α limit sets with regard to negative time and backward invariance.

Proof. (i). This follows from the definition of a flow and the definition of the alpha and the omega limit sets.

(ii). Assume $y \in \omega(U)$. By definition $y \in \overline{\varphi([T, \infty), U)}$ for all $T \geq 0$. In particular, for each $k \in \mathbb{Z}_+$ there exist $s_k \geq 0$, $x_k \in U$ with $y_k \stackrel{\text{def}}{=} \varphi(k + s_k, x_k) \in \varphi([k, \infty), U)$ such that $\|y_k - y\| \leq \frac{1}{k}$. Let $t_k \stackrel{\text{def}}{=} k + s_k$. Then $\lim_{k \rightarrow \infty} \varphi(t_k, x_k) = y$.

Conversely, assume there exists a sequence $t_k \rightarrow \infty$ and points $x_k \in U$ such that $\lim_{k \rightarrow \infty} \varphi(t_k, x_k) = y$. Then $y \in \overline{\varphi([t, \infty), U)}$ for all $t \geq 0$ and hence $y \in \omega(U)$.

The final remark follows from the observation that $\varphi([t, \infty), U) = \varphi(t, \varphi([0, \infty), U))$.

(iii). Since $U \subset V$ and V is forward invariant

$$\varphi([0, \infty), U) \subset \varphi([0, \infty), V) \subset V.$$

Since V is closed $\overline{\varphi([0, \infty), U)} \subset V$ and hence $\omega(U) \subset V$.

(iv). Since U is forward invariant

$$\varphi(t + s, U) \subset \varphi(t, \varphi(s, U)) \subset \varphi(t, U)$$

for all $t, s \geq 0$. Therefore, $\varphi([t, \infty), U) = \varphi(t, U)$, from which (5.3) follows.

(v). By definition, alpha and omega limit sets are defined in terms of intersections of closed sets and hence are closed.

We prove that $\omega(U)$ is forward and backward invariant, which implies that it is invariant. As a preliminary step we show that if U is forward invariant, then $\omega(U)$ is forward

invariant. Observe that by (iv) for all $t \geq 0$,

$$\begin{aligned}
 \varphi(t, \omega(U)) &= \varphi\left(t, \bigcap_{s \geq 0} \overline{\varphi(s, U)}\right) \\
 &\subset \bigcap_{s \geq 0} \varphi\left(t, \overline{\varphi(s, U)}\right) \\
 &\subset \bigcap_{s \geq 0} \overline{\varphi(t, \varphi(s, U))} \\
 &\subset \bigcap_{s \geq 0} \overline{\varphi(s, \varphi(t, U))} \\
 &= \omega(\varphi(t, U)) \\
 &= \omega(U)
 \end{aligned}$$

where the inclusions follow from the fact that φ is continuous and the last equality follows from (i). To finish the proof we note that $\varphi([0, \infty), U)$ is forward invariant. Thus

$$\varphi(t, \omega(U)) = \varphi(t, \varphi([0, \infty), U)) = \omega(\varphi([0, \infty), U)) = \omega(U)$$

where the last equality follows from (ii). This shows that $\omega(U)$ is forward invariant. A similar argument shows that U is backward invariant.

If U is a closed invariant set, then U is forward invariant. Thus by (iv)

$$U = \bigcap_{t \geq 0} \varphi(t, U) \subset \bigcap_{t \geq 0} \overline{\varphi(t, U)} = \omega(U).$$

However, since U is a closed forward invariant set by (ii), $\omega(U) \subset U$.

(vi). Since for every $T \geq 0$, $\varphi([T, \infty), U) \subset \varphi([0, \infty), U) \subset K$ and K is compact, then $\overline{\varphi([T, \infty), U)} \subset K$ is compact. By definition this implies that $\omega(U)$ is the intersection of a nested collection of non empty compact sets and so is nonempty and compact.

Now assume that U is connected. Then $\varphi([T, \infty), U)$ is connected and thus $\overline{\varphi([T, \infty), U)}$ is connected. Again, since $\omega(U)$ is the intersection of a nested collection of compact connected sets, it is connected.

(vii). Since $\omega(U)$ is a closed invariant set, by (iii) $\omega(V) \subset \omega(U)$. The corresponding result of 3 for alpha limit sets shows $\alpha(V) \subset \alpha(U)$. $\square$

It is important to observe that $\alpha(U)$ and $\omega(U)$ only describe the asymptotic dynamics of points in U and ignores that of nearby points. Returning to Example 5.1.8 observe that $\omega(0) = 0$, but $\omega(y) = \{x \in \mathbb{R}^2 \mid \|x\| = |K|\}$ for all $y \in \mathbb{R}^2 \setminus \{0\}$. To discuss sets whose asymptotic dynamics is in agreement with that of its neighbors we introduce the following concepts.

Definition 5.1.11. Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow. A forward invariant set U is a *trapping region* if there exists $T > 0$ such that

$$\overline{\varphi(T, U)} \subset \text{int}(U).$$

Dually, a backward invariant set U is a *repelling region* if there exists $T < 0$ such that

$$\overline{\varphi(T, U)} \subset \text{int}(U).$$

Definition 5.1.12. Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow on a locally compact metric space. A set $A \subset X$ is an *attractor* if there exists a trapping region U such that $A = \omega(U)$. A set $R \subset X$ is a *repeller* if there exists a repelling region U such that $R = \alpha(U)$.

Definition 5.1.13. Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow on a locally compact metric space. The *maximal invariant set* in U is defined by

$$\text{Inv}(U, \varphi) \stackrel{\text{def}}{=} \{x \in U \mid \varphi(\mathbb{R}, x) \subset U\}.$$

We leave the proof of the following proposition as an exercise.

Proposition 5.1.14. *If U is a trapping region and $A = \omega(U)$, then A is the maximal invariant set in U .*

In applications it can be difficult to determine a trapping region, because they must be forward invariant. A seemingly weaker notion is the following.

Definition 5.1.15. Let $\varphi: \mathbb{R} \times X \rightarrow X$ be a flow on a locally compact metric space. A compact set N is called an *attracting neighborhood* if

$$\omega(N) \subset \text{int}(N).$$

Dually, a compact set N is a *repelling neighborhood* if $\alpha(N) \subset \text{int}(N)$.

In general attracting neighborhoods are easier to identify than trapping regions (see Exercise 4.7.4). However, as the following results indicate they are closely related.

Proposition 5.1.16. *If U is a trapping region, then $\overline{U}$ is an attracting neighborhood. If N is an attracting neighborhood, then there exists a trapping region U such that $\omega(N) \subset \overline{U} \subset N$.*

The proof of Proposition 5.1.16 follows from the continuity of the flow. However, the argument is somewhat technical thus we do not present it here.

Corollary 5.1.17. *A is an attractor for a flow if and only if there exists an attracting neighborhood N such that $A = \omega(N)$.*

The proof of the following result is nontrivial.

Proposition 5.1.18. *Let $X \subset \mathbb{R}^n$ be a compact neighborhood of $\tilde{x}$. An equilibrium $\tilde{x}$ is an attractor for the flow $\varphi: \mathbb{R} \times X \rightarrow X$ if and only if $\tilde{x}$ is an asymptotically stable equilibrium.*

5.2 Gradient Systems

Definition 5.2.1. A *gradient system* on an open set $U \subset \mathbb{R}^n$ is a differential equation of the form

$$\dot{x} = -\nabla V(x) \quad (5.4)$$

where $V: U \rightarrow \mathbb{R}$ is a C^2 function.

Theorem 5.2.2. Given a gradient system $\dot{x} = -\nabla V(x)$ defined on $U \subset \mathbb{R}^n$

$$\frac{d}{dt}V(x) \leq 0$$

for all $x \in U$. Furthermore,

$$\frac{d}{dt}V(x) = 0 \quad \text{if and only if} \quad x \text{ is an equilibrium of the system.}$$

Proof. Observe that

$$\begin{aligned} \frac{d}{dt}V(x) &= \nabla V(x) \cdot \dot{x} \\ &= \nabla V(x) \cdot -\nabla V(x) \\ &= -\|\nabla V(x)\|^2 \leq 0 \end{aligned}$$

Furthermore, we obtain equality if and only if $\nabla V(x) = 0$ in which case x is an equilibrium. $\square$

We wish to prove the following result.

Proposition 5.2.3. Consider a gradient system $\dot{x} = -\nabla V(x)$. If $\tilde{x}$ be an isolated minimum of V , then $\tilde{x}$ is an asymptotically stable equilibrium.

We begin with a slightly more general result.

Theorem 5.2.4. Consider the differential equation $\dot{x} = f(x)$ with an equilibrium point $\tilde{x}$. Let U be a neighborhood of $\tilde{x}$. Let $V: U \rightarrow \mathbb{R}^n$ be a continuous function that is differentiable on $U \setminus \tilde{x}$. If

(i) $V(x) \geq 0$ and $V(x) = 0$ if and only if $x = \tilde{x}$, and

(ii) $\frac{d}{dt}V(x) \leq 0$ in $U \setminus \tilde{x}$,

then $\tilde{x}$ is stable. Furthermore, if in addition

(iii) $\frac{d}{dt}V(x) < 0$ in $U \setminus \tilde{x}$,

then $\tilde{x}$ is asymptotically stable.

Proof. □

Traditionally, a function V satisfying (i) and (ii) is called a *Lyapunov function* for $\tilde{x}$.

Theorem 5.2.5. *Consider a gradient system $\dot{x} = f(x)$ with equilibrium $\tilde{x}$. Then the eigenvalues of $Df(\tilde{x})$ are real.*

Proof. □

Discuss limits of solutions.

Discuss the structure of $S(a, b) := \{x \in \mathbb{R}^n \mid \varphi(\mathbb{R}, x) \subset V^{-1}([a, b])\}$.

5.3 Lyapunov functions

5.4 Hamiltonian Systems

Definition 5.4.1. A *Hamiltonian system* is a $2n$ dimensional differential equation of the form

$$\dot{q}_i = \frac{\partial}{\partial p_i} H(t, q, p), \quad \dot{p}_i = -\frac{\partial}{\partial q_i} H(t, q, p), \quad i = 1, \dots, n \quad (5.5)$$

where the *Hamiltonian function* $H: \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable.

Often written as

$$\dot{z} = J \nabla H(t, z)$$

where J is the $2n \times 2n$ skew symmetric matrix

$$J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}.$$

Theorem 5.4.2. *Given a Hamiltonian system $\dot{z} = J \nabla H(z)$ defined on $U \subset \mathbb{R}^{2n}$*

$$\frac{d}{dt} H(z) = 0$$

for all $z \in U$.

Proof. Observe that

$$\begin{aligned} \frac{d}{dt} H(q, p) &= \sum_{i=1}^n \frac{\partial H}{\partial q_i} \frac{dq_i}{dt} + \sum_{i=1}^n \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} \\ &= 0 \end{aligned}$$

□

Theorem 5.4.3 (Liouville). *Consider a Hamiltonian system $\dot{z} = J\nabla H(t, z)$ and a bounded set $U \subset \mathbb{R}^{2n}$. Then, the volume of U is preserved under the flow.*

Proof. □

Theorem 5.4.4 (Poincaré Recurrence). *Consider a Hamiltonian system $\dot{z} = J\nabla H(t, z)$. Let $S \subset \mathbb{R}^{2n}$ be a bounded region that is an invariant set. Given any open set $U \subset S$, there exists $z \in U$ such that $\varphi(n, z) \in U$ for some integer $n \geq 1$.*

Proof. Let $U_n := \varphi(n, U)$. Each set has the same volume, but $U_n \subset S$ for all $n \in \mathbb{Z}$. Therefore, there some of the sets must intersect. Assume $U_{n_0} \cap U_{n_1} \neq \emptyset$ and $n_0 < n_1$. Observe that $U = \varphi(-n_0, U_{n_0})$ and hence $U \cap \varphi(n_1 - n_0, U_{n_1}) \neq \emptyset$. □

5.5 Exercises